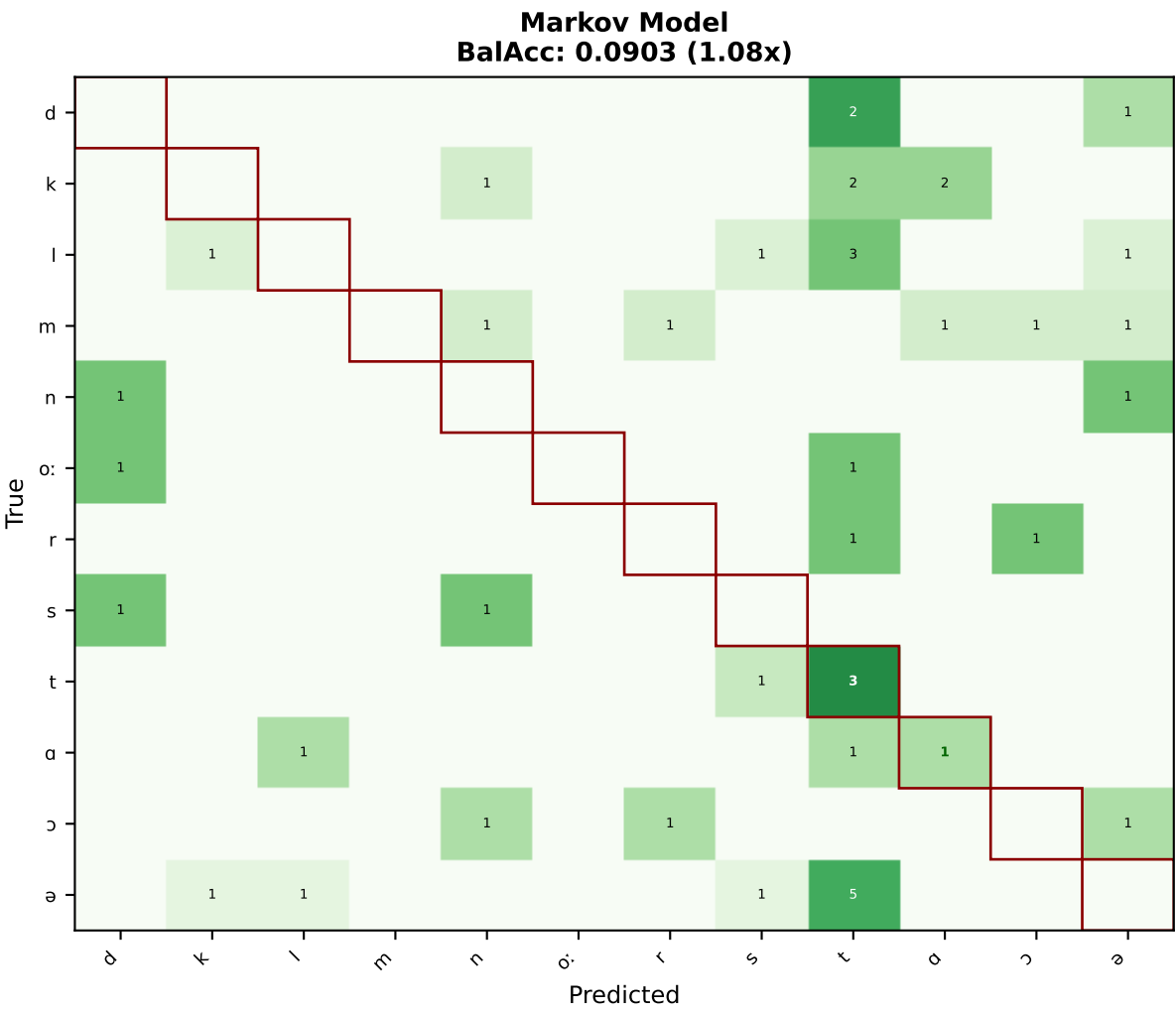
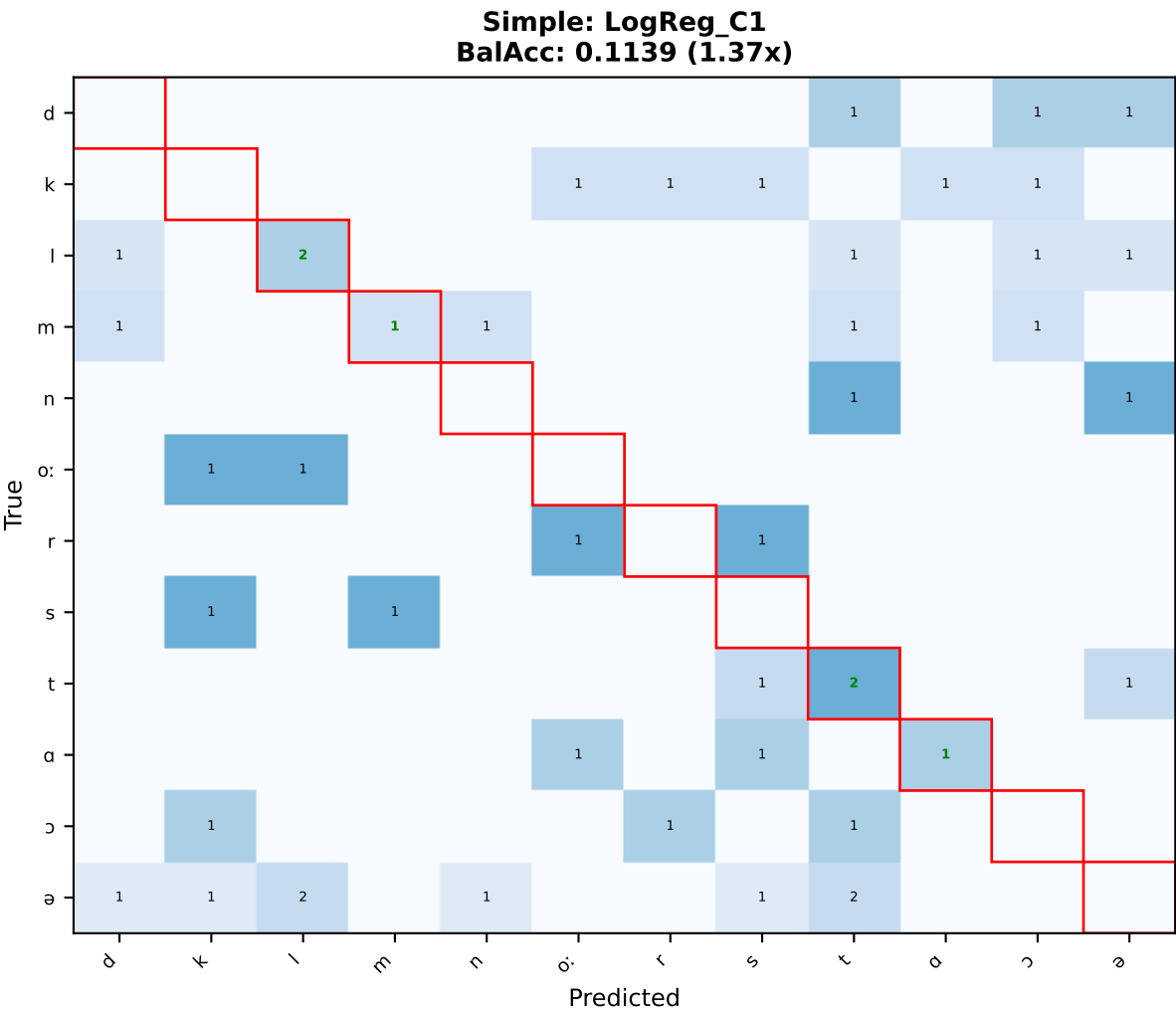


Pipeline Optimization Report - Simple vs Markov Comparison

Patient	Classifier	Simple Acc	Simple Lift	Markov Acc	Markov Lift	Winner	Diff
P01	LogReg_C1	0.114	1.37x	0.090	1.08x	Simple	-0.024
P02	KNN_15	0.141	1.55x	0.159	1.75x	Markov	+0.018
P04	LogReg_C1	0.086	1.03x	0.113	1.36x	Markov	+0.027
P06	SVM_RBF_C1	0.114	1.26x	0.076	0.83x	Simple	-0.038
P07	KNN_35	0.129	1.42x	0.197	2.17x	Markov	+0.068
P09	EasyEnsemble	0.037	0.37x	0.117	1.17x	Markov	+0.080
P10	LogReg_C0.01	0.217	1.52x	0.161	1.13x	Simple	-0.056
P17	LogReg_C1	0.160	0.80x	0.180	0.90x	Markov	+0.020
P21	LogReg_C1	0.029	0.72x	0.026	0.65x	Simple	-0.003
P22	SVM_RBF_C1	0.043	1.21x	0.040	1.11x	Simple	-0.004
P24	AdaBoost_100	0.037	1.08x	0.034	0.97x	Simple	-0.004
P26	LogReg_C0.01	0.019	0.50x	0.049	1.29x	Markov	+0.030
P28	LogReg_C0.1	0.078	1.94x	0.030	0.74x	Simple	-0.048
P29	LogReg_C1	0.018	0.52x	0.042	1.22x	Markov	+0.024
P30	RF_balance..	0.042	1.29x	0.035	1.08x	Simple	-0.007

Patient: P01 | Winner: Simple | Simple: 0.114 vs Markov: 0.090

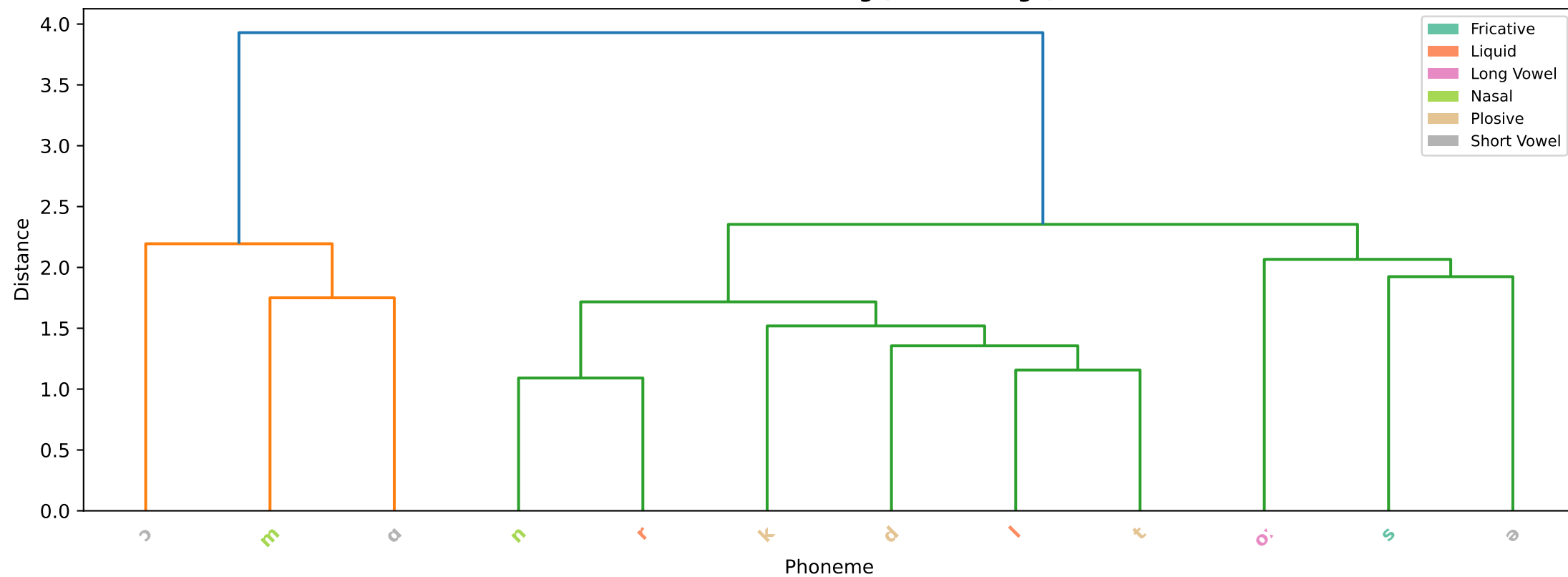


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
d	3	0.00	0.00	0.00	0.00	0.00	0.00	-
k	5	0.00	0.00	0.00	0.00	0.00	0.00	-
l	6	0.33	0.40	0.36	0.00	0.00	0.00	S
m	5	0.20	0.50	0.29	0.00	0.00	0.00	S
n	2	0.00	0.00	0.00	0.00	0.00	0.00	-
o:	2	0.00	0.00	0.00	0.00	0.00	0.00	-
r	2	0.00	0.00	0.00	0.00	0.00	0.00	-
s	2	0.00	0.00	0.00	0.00	0.00	0.00	-
t	4	0.50	0.22	0.31	0.75	0.17	0.27	S
a	3	0.33	0.50	0.40	0.33	0.25	0.29	S
ɔ	3	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	8	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	45	0.11	0.14	0.11	0.09	0.03	0.05	S

Phoneme Clustering Analysis - P01

Hierarchical Clustering (Ward linkage)



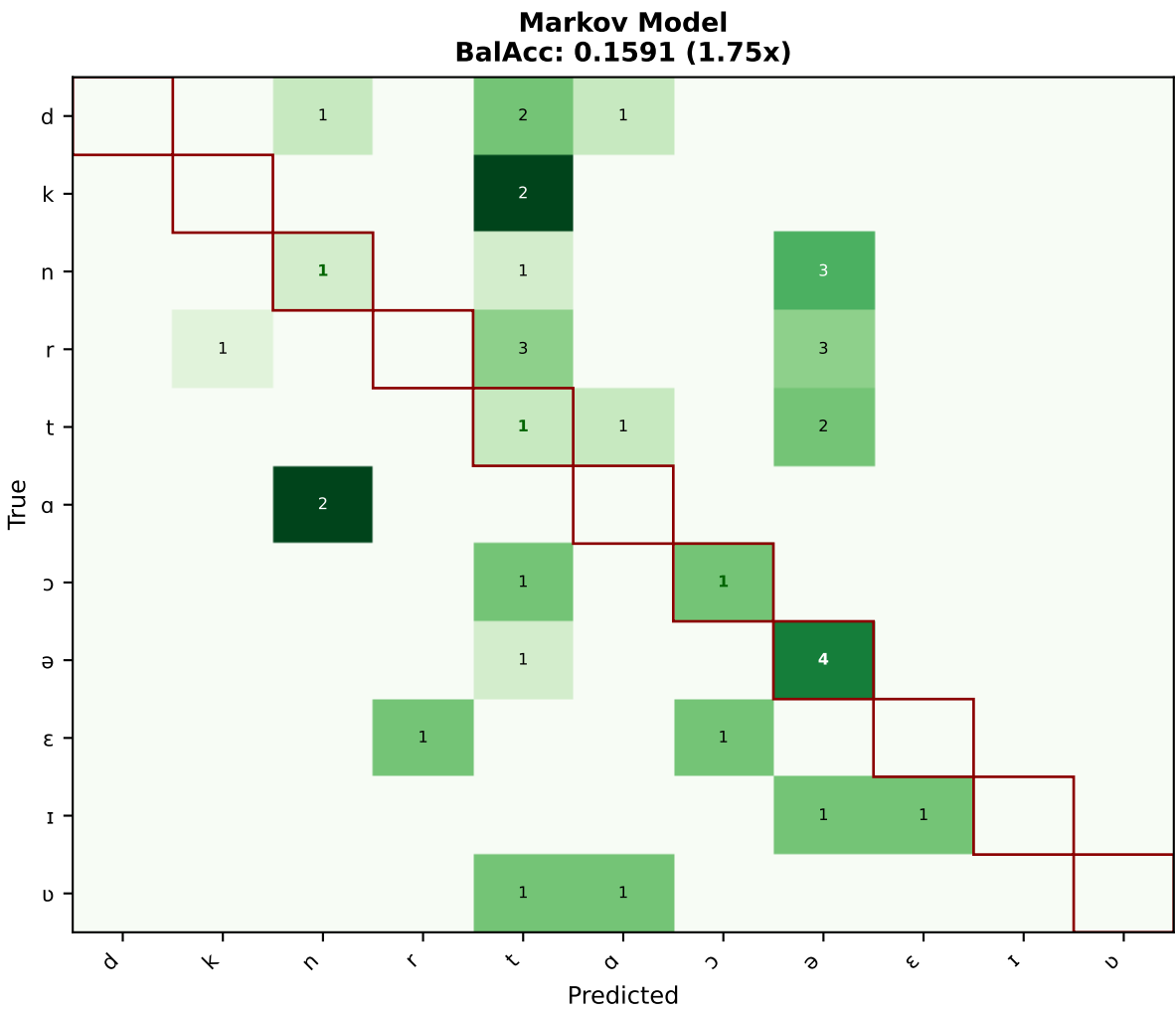
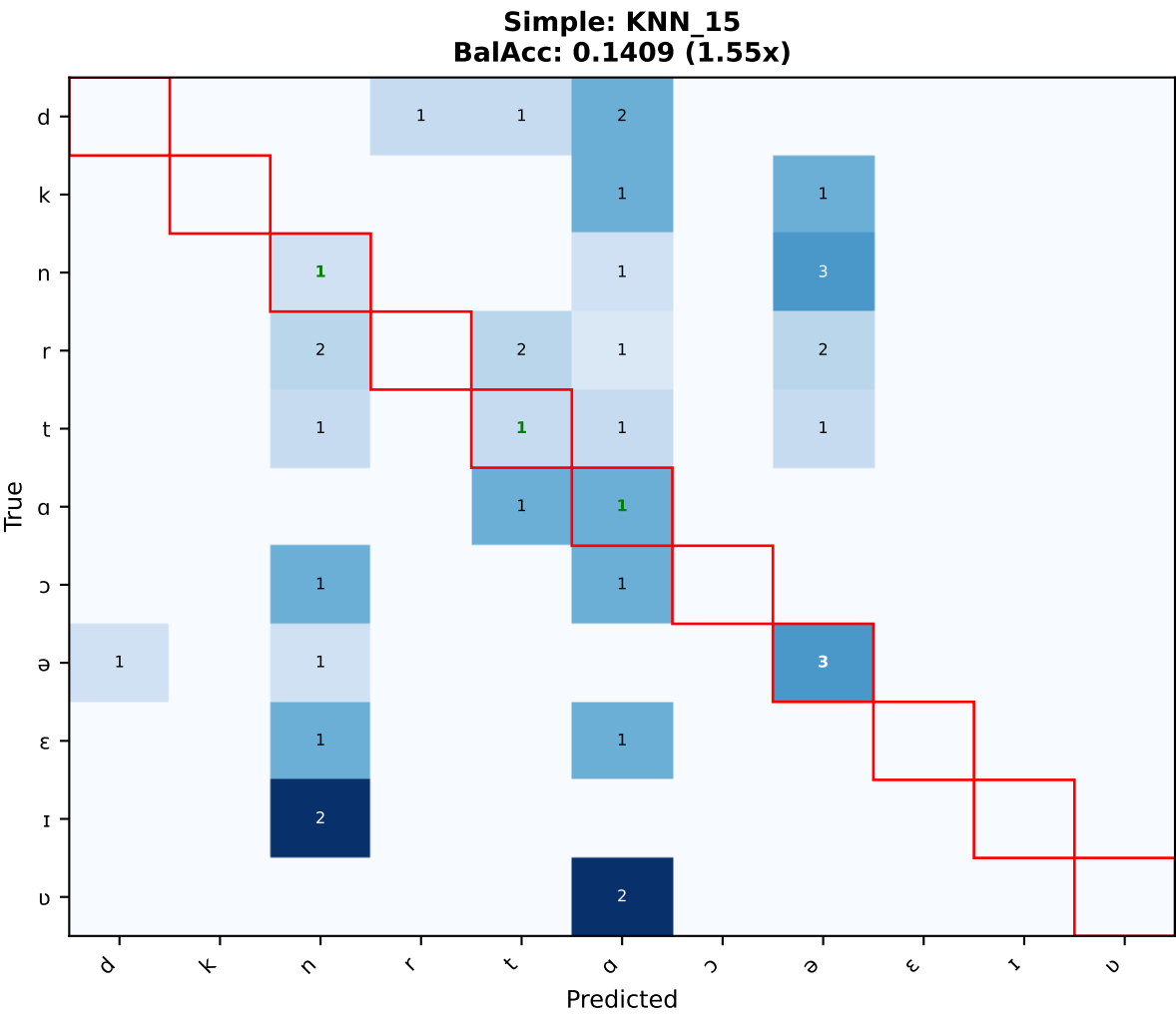
Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	2	29	Nasal	m, ɑ
Group 2	1	10	Short Vowel	ɔ
Group 3	6	146	Plosive	d, k, l, n, r, t
Group 4	2	61	Short Vowel	s, ə
Group 5	1	10	Long Vowel	o:

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	1	20	s
Liquid	2	43	l, r
Long Vowel	1	10	o:
Nasal	2	41	m, n
Plosive	3	73	d, k, t
Short Vowel	3	69	ɑ, ɔ, ə

Patient: P02 | Winner: Markov | Simple: 0.141 vs Markov: 0.159

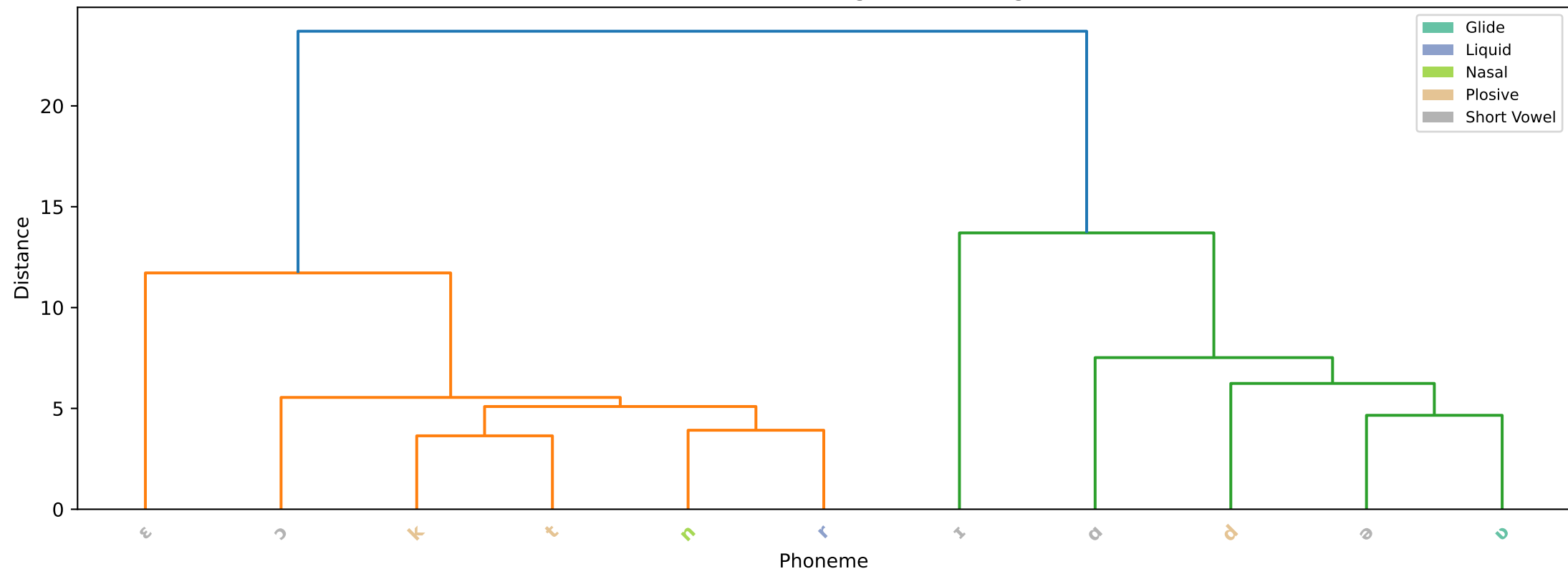


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
d	4	0.00	0.00	0.00	0.00	0.00	0.00	-
k	2	0.00	0.00	0.00	0.00	0.00	0.00	-
n	5	0.20	0.11	0.14	0.20	0.25	0.22	M
r	7	0.00	0.00	0.00	0.00	0.00	0.00	-
t	4	0.25	0.20	0.22	0.25	0.08	0.12	S
a	2	0.50	0.09	0.15	0.00	0.00	0.00	S
ɔ	2	0.00	0.00	0.00	0.50	0.50	0.50	M
ə	5	0.60	0.30	0.40	0.80	0.31	0.44	M
ɛ	2	0.00	0.00	0.00	0.00	0.00	0.00	-
ɪ	2	0.00	0.00	0.00	0.00	0.00	0.00	-
ʊ	2	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	37	0.14	0.06	0.08	0.16	0.10	0.12	M

Phoneme Clustering Analysis - P02

Hierarchical Clustering (Ward linkage)



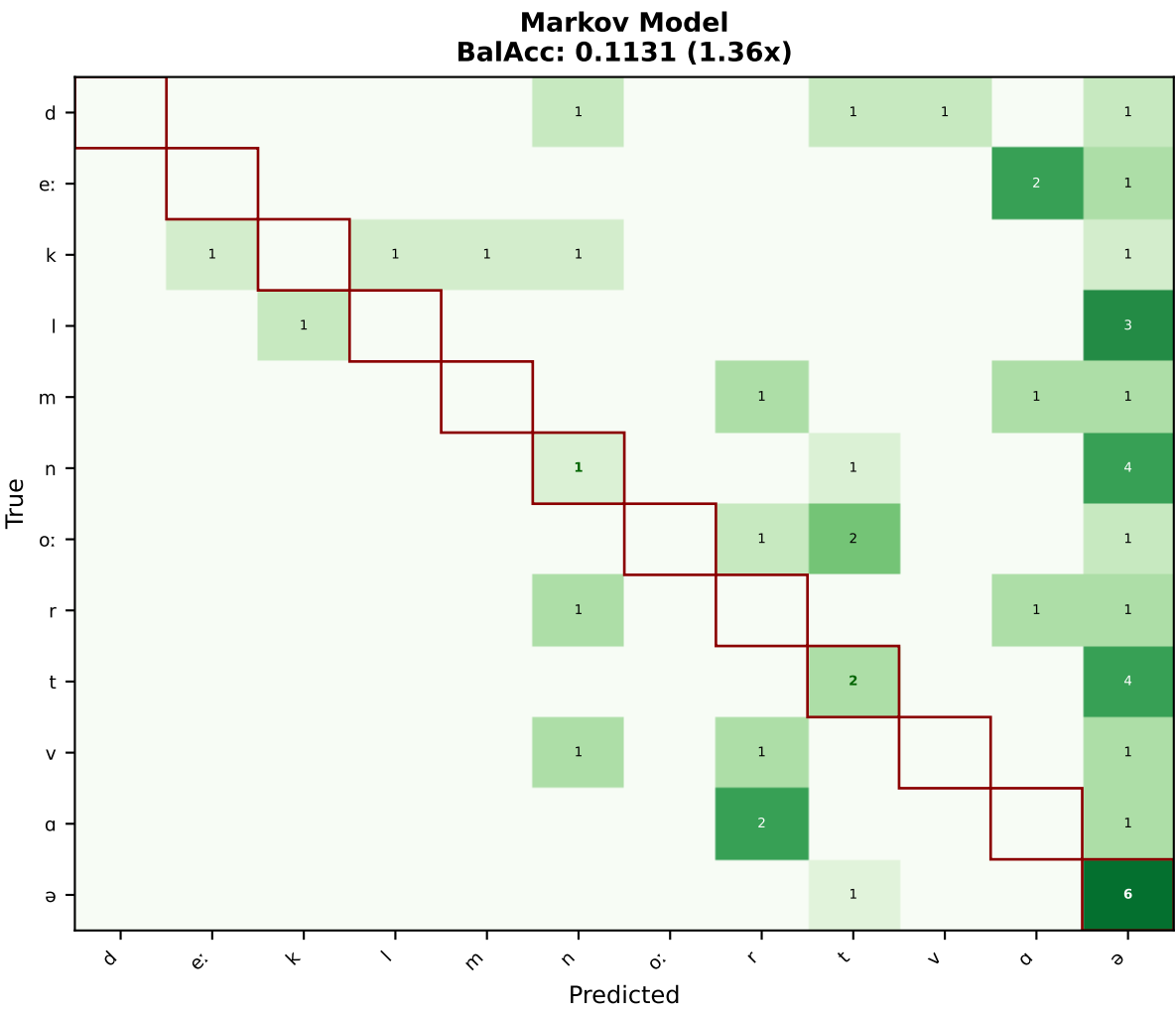
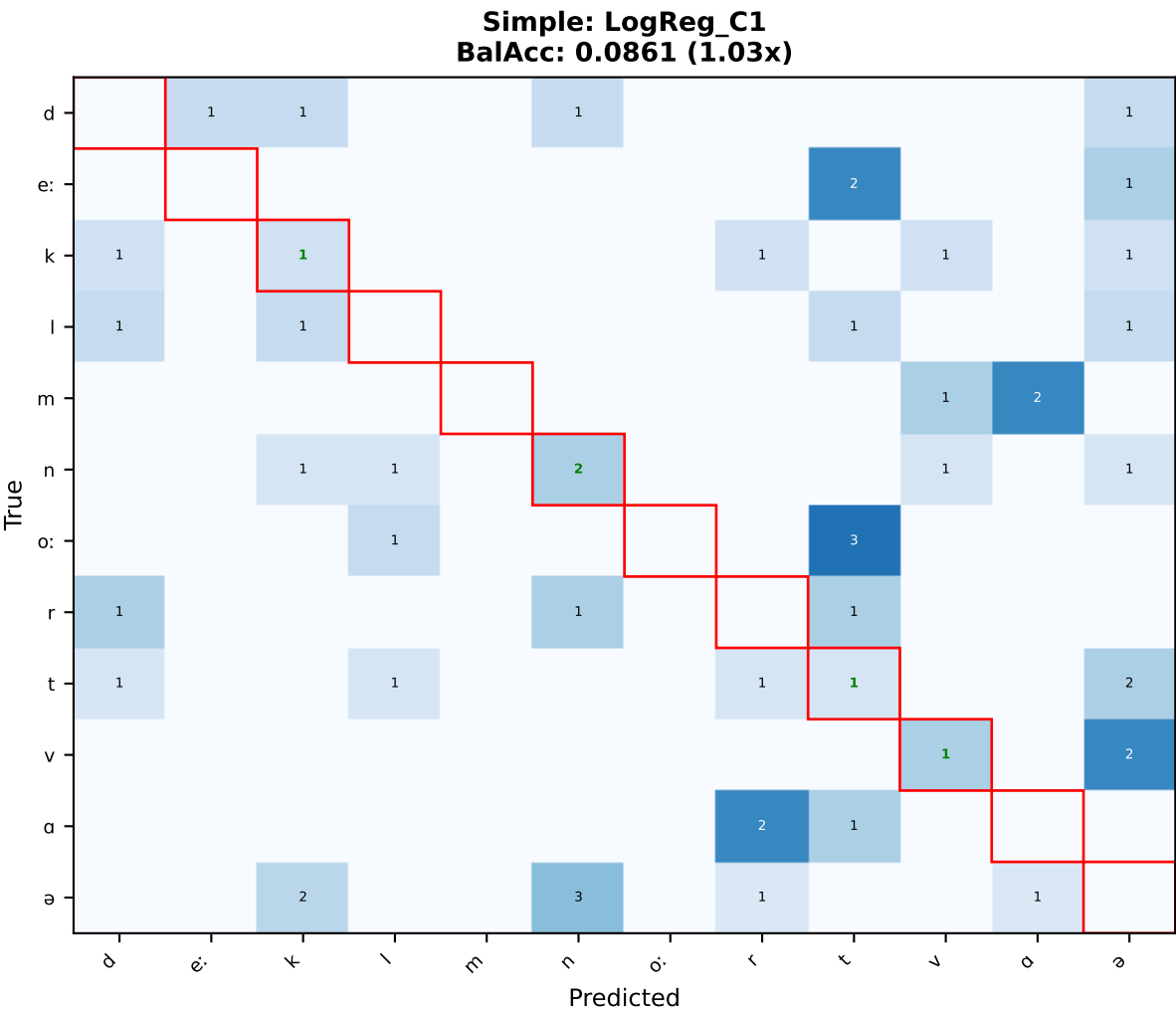
Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	5	133	Plosive	k, n, r, t, ɔ
Group 2	1	10	Short Vowel	ε
Group 3	3	71	Plosive	d, ə, ʊ
Group 4	1	22	Short Vowel	ɑ
Group 5	1	9	Short Vowel	ɪ

Phonetic Categories

Category	N	Samples	Phonemes
Glide	1	12	ʊ
Liquid	1	25	r
Nasal	1	38	n
Plosive	3	75	d, k, t
Short Vowel	5	95	ɑ, ɔ, ə, ε, ɪ

Patient: P04 | Winner: Markov | Simple: 0.086 vs Markov: 0.113

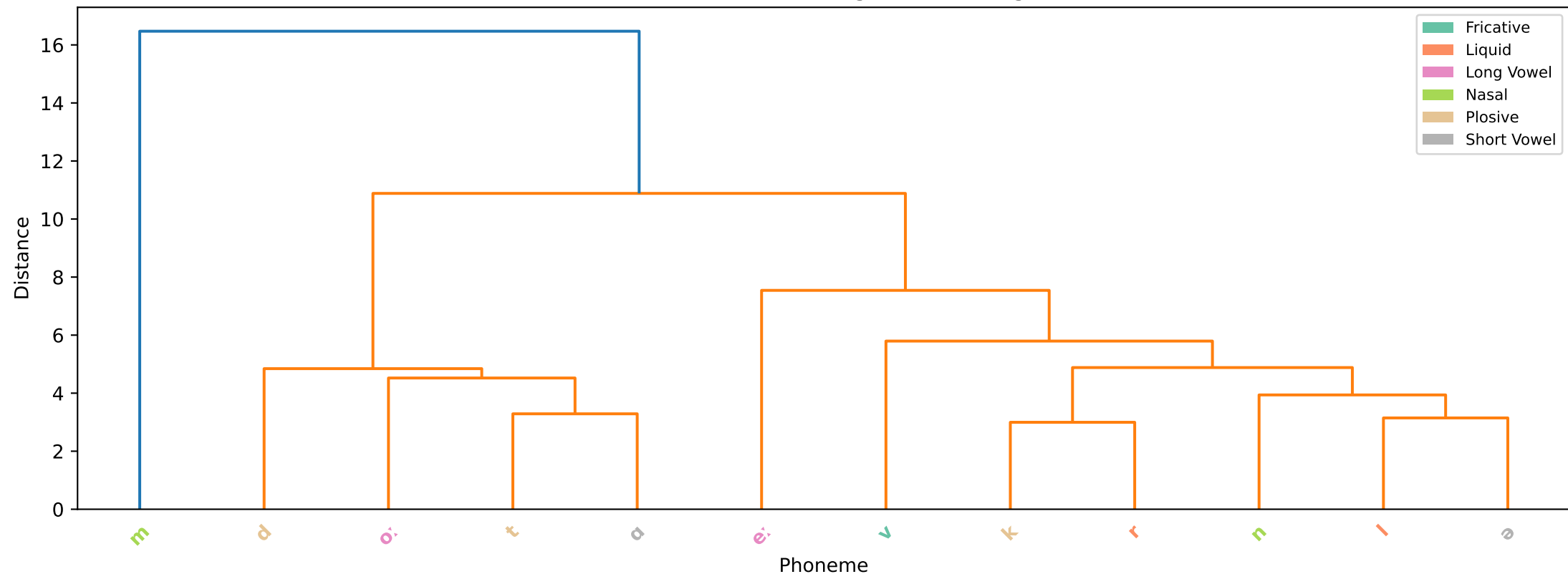


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
d	4	0.00	0.00	0.00	0.00	0.00	0.00	-
e:	3	0.00	0.00	0.00	0.00	0.00	0.00	-
k	5	0.20	0.17	0.18	0.00	0.00	0.00	S
l	4	0.00	0.00	0.00	0.00	0.00	0.00	-
m	3	0.00	0.00	0.00	0.00	0.00	0.00	-
n	6	0.33	0.29	0.31	0.17	0.20	0.18	S
o:	4	0.00	0.00	0.00	0.00	0.00	0.00	-
r	3	0.00	0.00	0.00	0.00	0.00	0.00	-
t	6	0.17	0.11	0.13	0.33	0.29	0.31	M
v	3	0.33	0.25	0.29	0.00	0.00	0.00	S
a	3	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	7	0.00	0.00	0.00	0.86	0.24	0.38	M
TOTAL	51	0.09	0.07	0.08	0.11	0.06	0.07	M

Phoneme Clustering Analysis - P04

Hierarchical Clustering (Ward linkage)

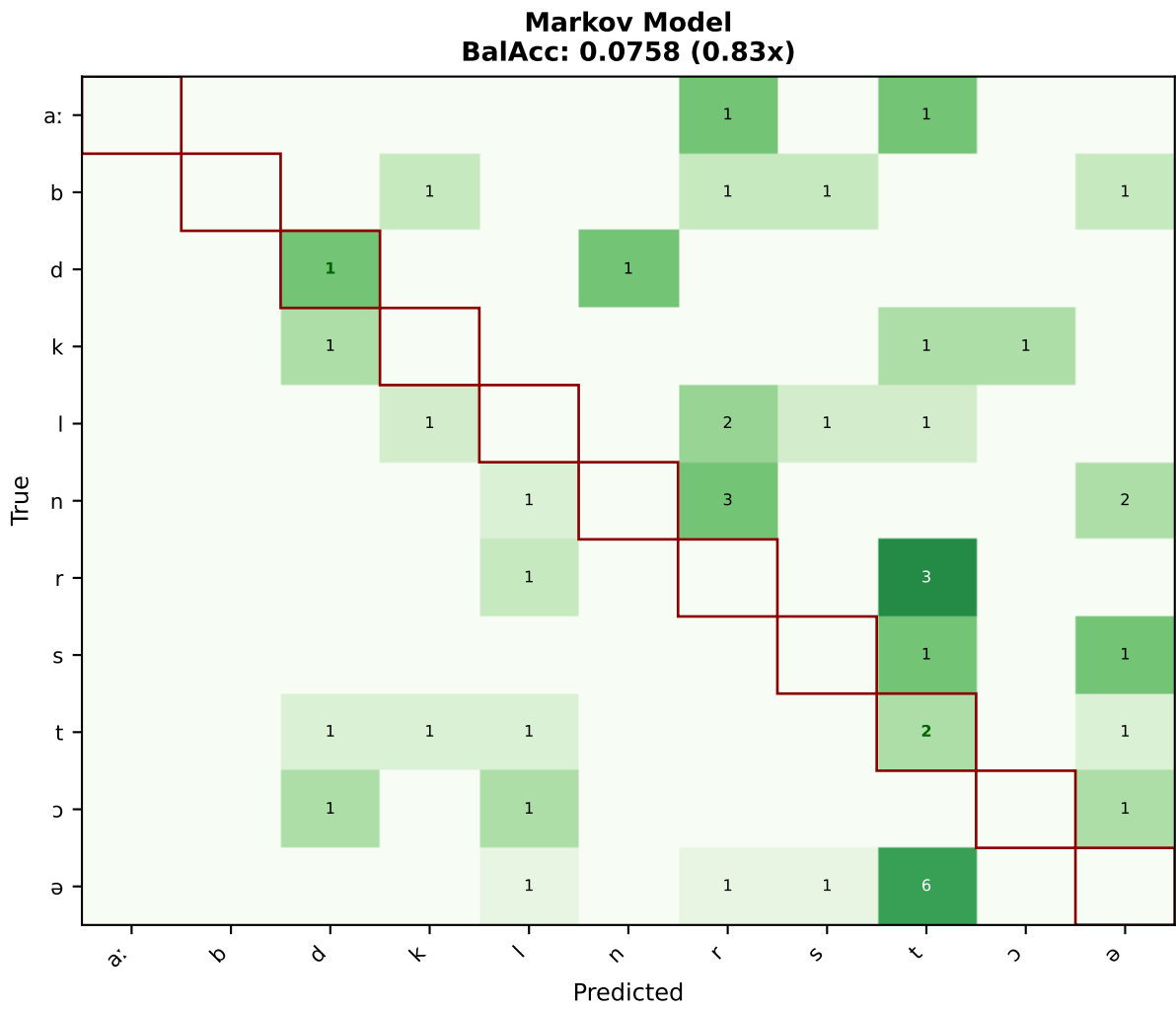
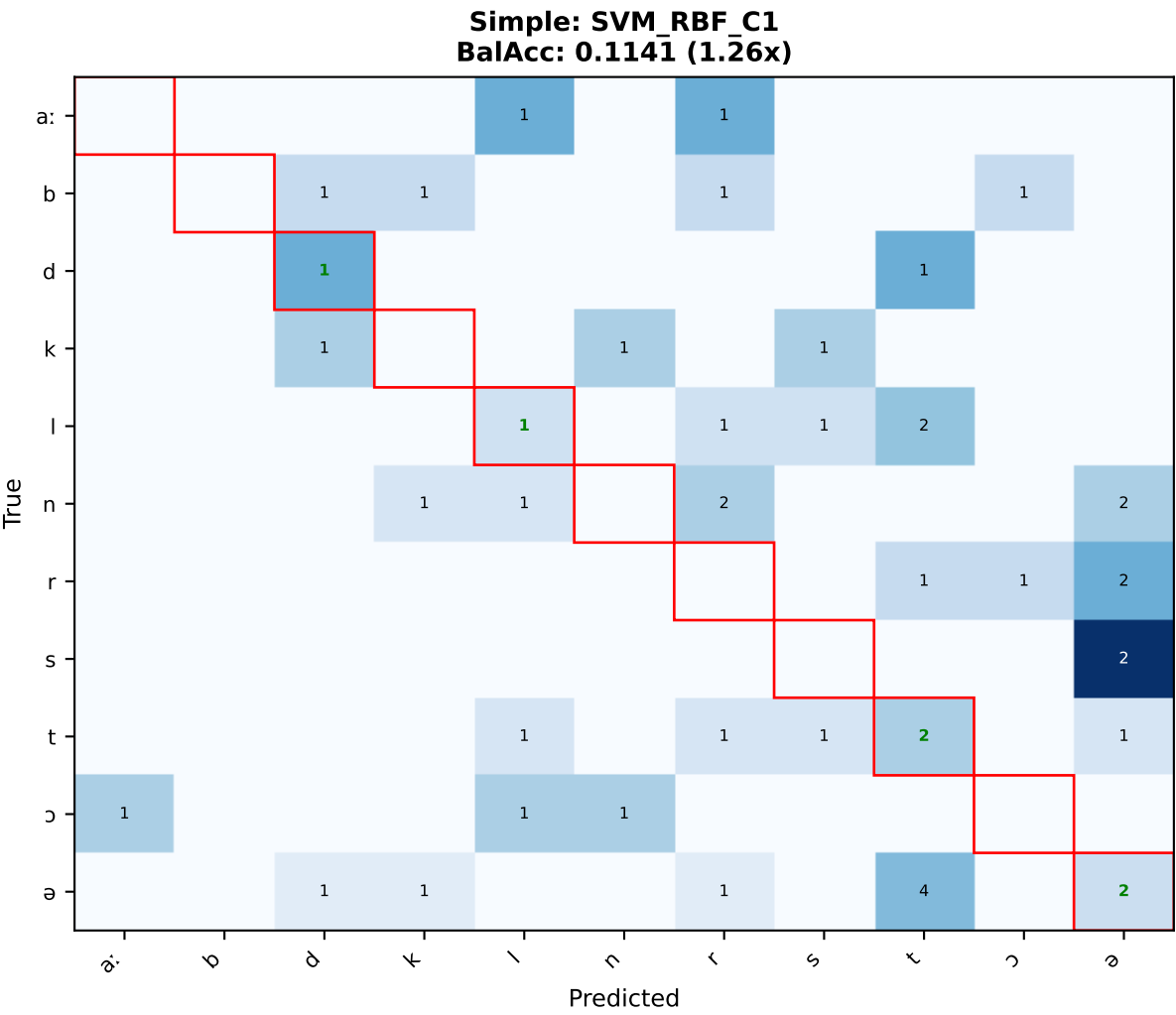


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	4	79	Plosive	d, o:, t, a
Group 2	5	125	Liquid	k, l, n, r, ə
Group 3	1	11	Fricative	v
Group 4	1	8	Long Vowel	e:
Group 5	1	11	Nasal	m

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	1	11	v
Liquid	2	45	l, r
Long Vowel	2	15	e:, o:
Nasal	2	36	m, n
Plosive	3	69	d, k, t
Short Vowel	2	58	a, ə

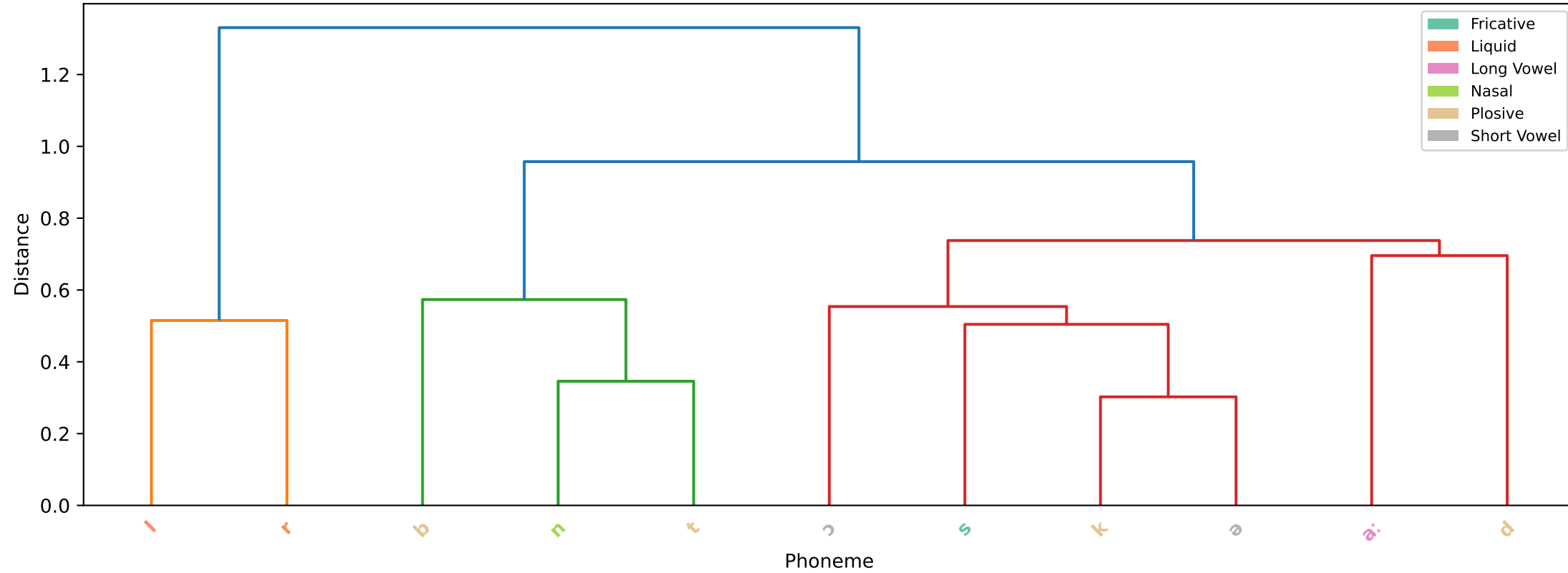


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
a:	2	0.00	0.00	0.00	0.00	0.00	0.00	-
b	4	0.00	0.00	0.00	0.00	0.00	0.00	-
d	2	0.50	0.25	0.33	0.50	0.25	0.33	-
k	3	0.00	0.00	0.00	0.00	0.00	0.00	-
l	5	0.20	0.20	0.20	0.00	0.00	0.00	S
n	6	0.00	0.00	0.00	0.00	0.00	0.00	-
r	4	0.00	0.00	0.00	0.00	0.00	0.00	-
s	2	0.00	0.00	0.00	0.00	0.00	0.00	-
t	6	0.33	0.20	0.25	0.33	0.13	0.19	S
ɔ	3	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	9	0.22	0.22	0.22	0.00	0.00	0.00	S
TOTAL	46	0.11	0.08	0.09	0.08	0.03	0.05	S

Phoneme Clustering Analysis - P06

Hierarchical Clustering (Ward linkage)

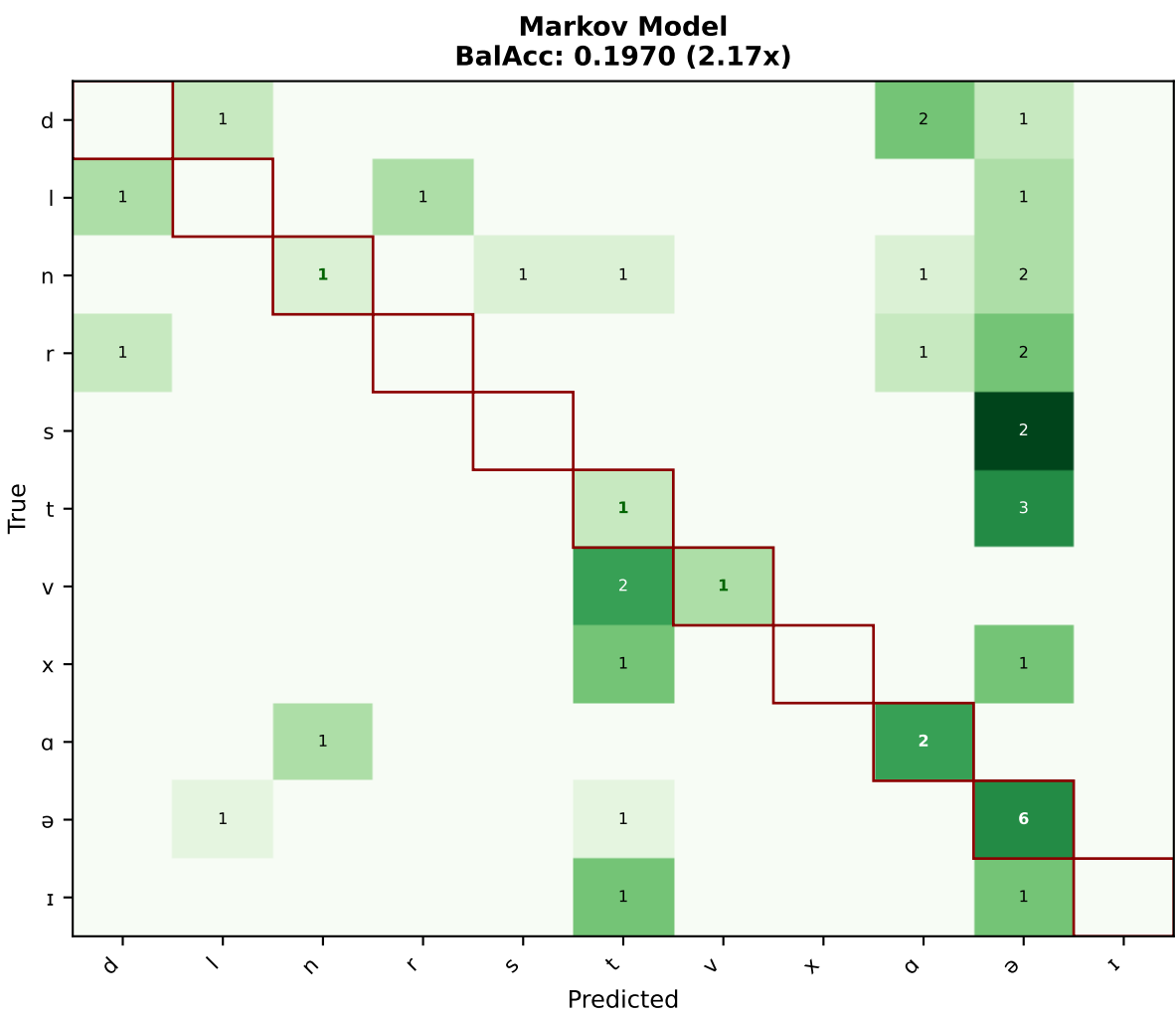
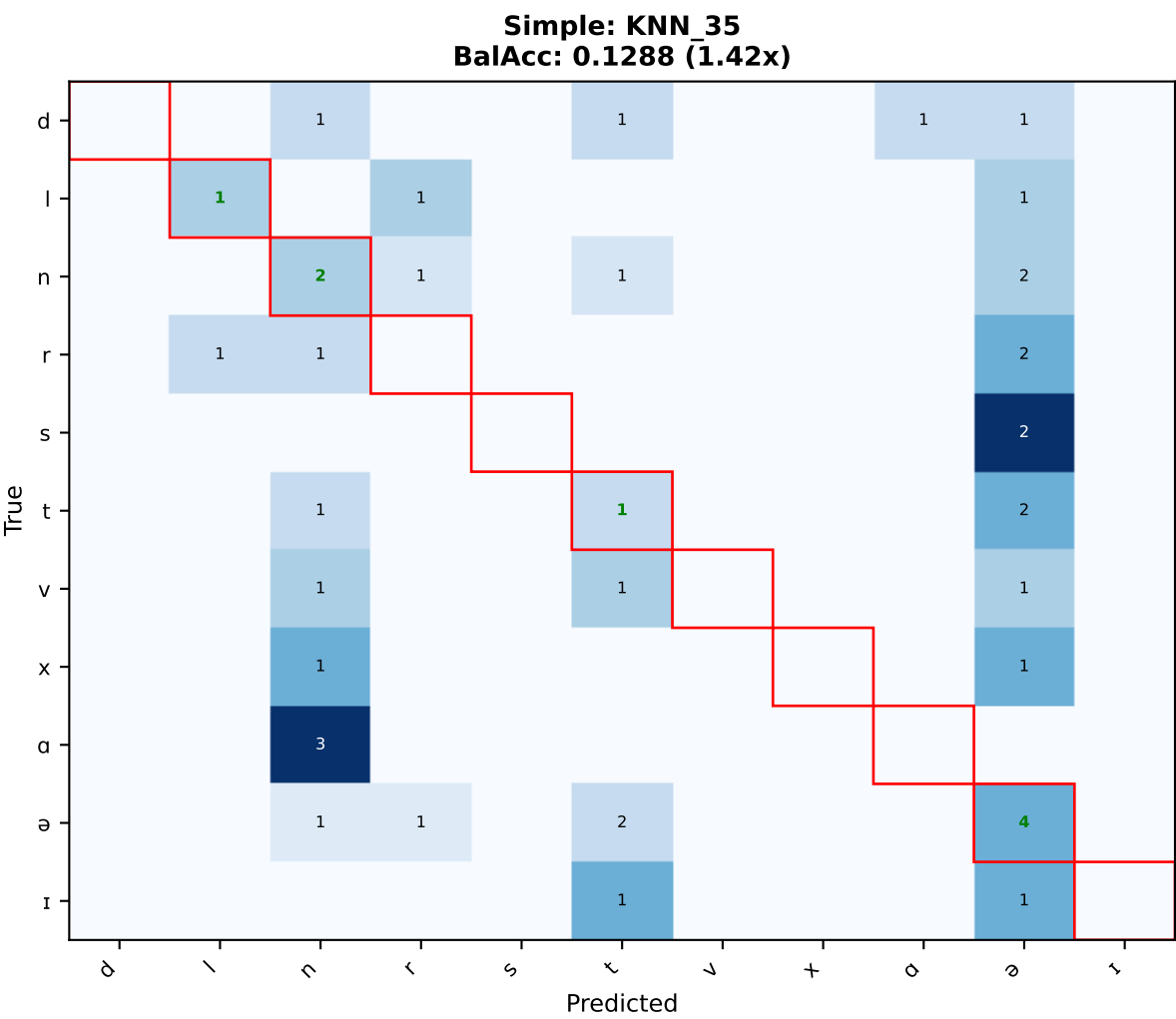


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	2	46	Liquid	l, r
Group 2	3	80	Plosive	b, n, t
Group 3	4	84	Short Vowel	k, s, ʊ, ə
Group 4	1	9	Long Vowel	a:
Group 5	1	18	Plosive	d

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	1	18	s
Liquid	2	46	l, r
Long Vowel	1	9	a:
Nasal	1	31	n
Plosive	4	84	b, d, k, t
Short Vowel	2	49	ʊ, ə

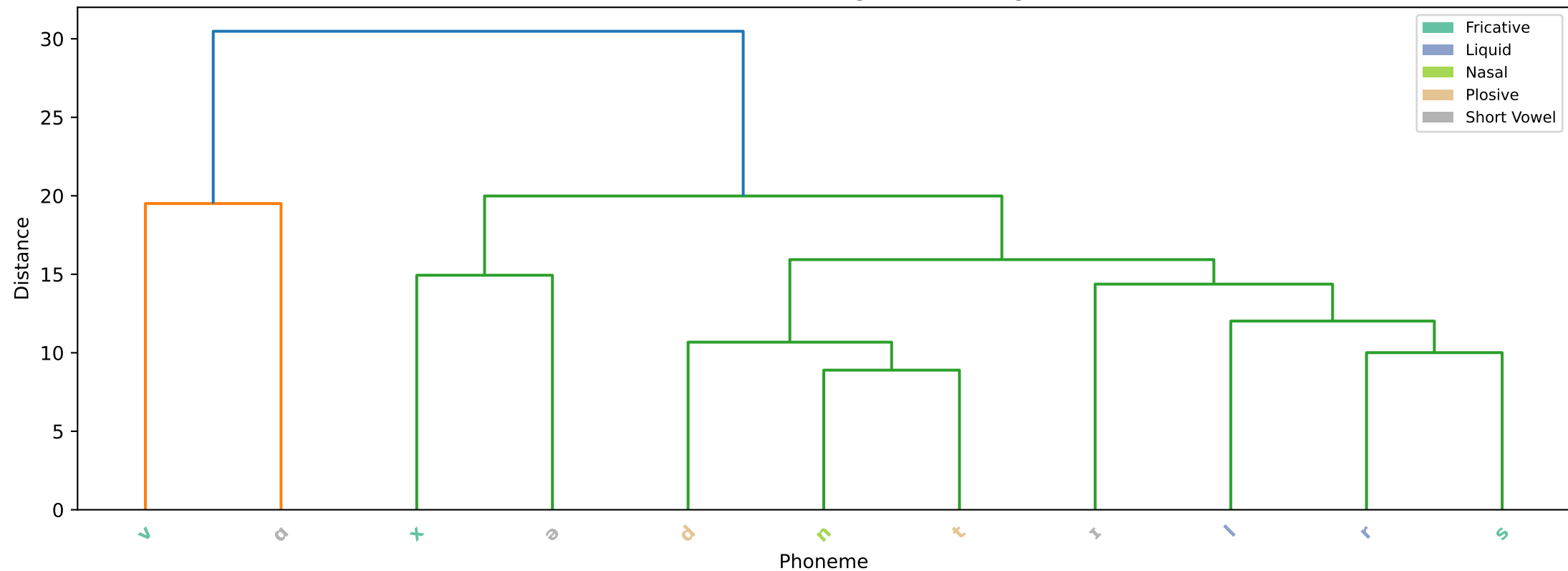


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
d	4	0.00	0.00	0.00	0.00	0.00	0.00	-
l	3	0.33	0.50	0.40	0.00	0.00	0.00	S
n	6	0.33	0.18	0.24	0.17	0.50	0.25	M
r	4	0.00	0.00	0.00	0.00	0.00	0.00	-
s	2	0.00	0.00	0.00	0.00	0.00	0.00	-
t	4	0.25	0.14	0.18	0.25	0.14	0.18	-
v	3	0.00	0.00	0.00	0.33	1.00	0.50	M
x	2	0.00	0.00	0.00	0.00	0.00	0.00	-
a	3	0.00	0.00	0.00	0.67	0.33	0.44	M
ə	8	0.50	0.24	0.32	0.75	0.32	0.44	M
i	2	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	41	0.13	0.10	0.10	0.20	0.21	0.17	M

Phoneme Clustering Analysis - P07

Hierarchical Clustering (Ward linkage)

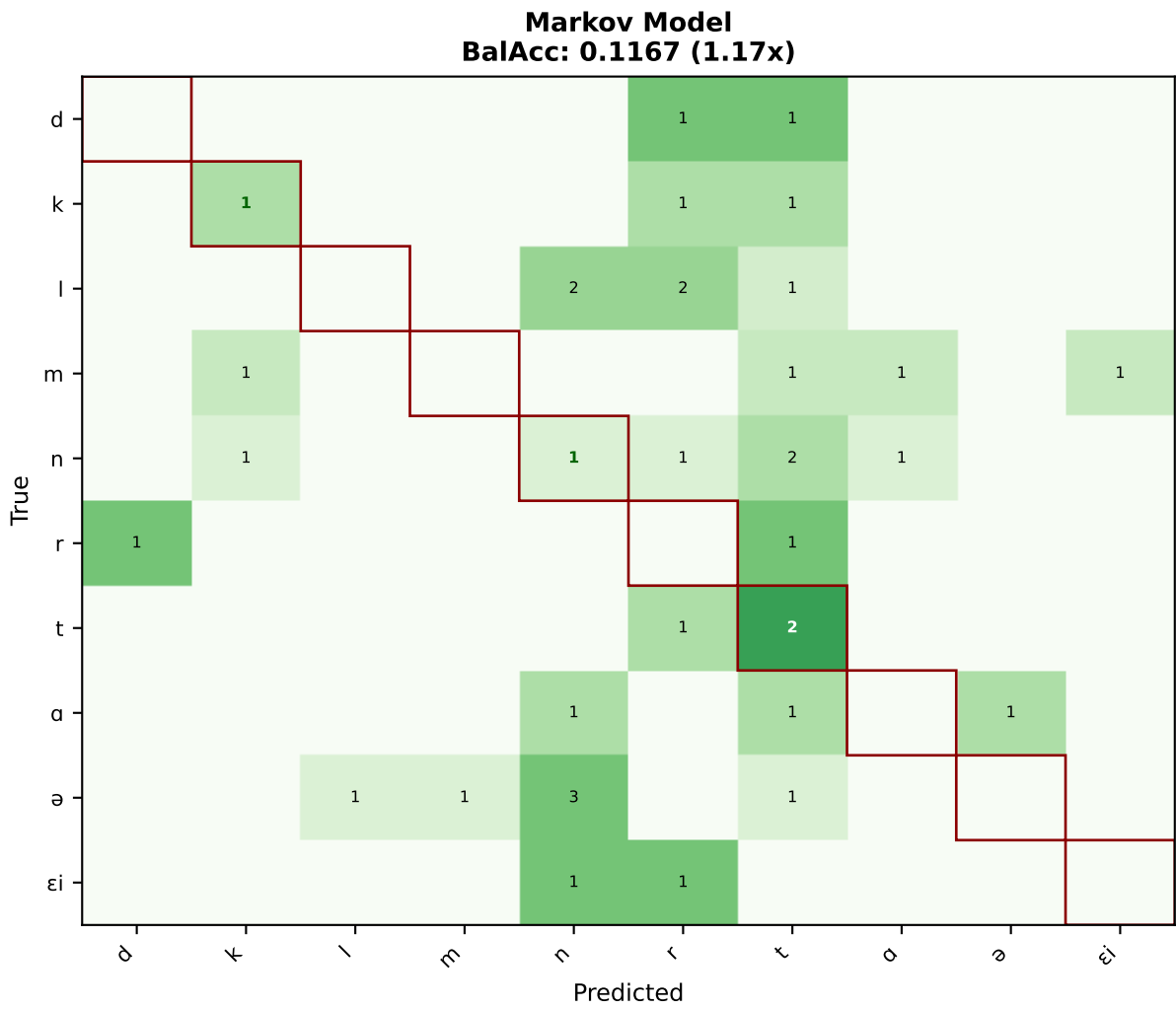
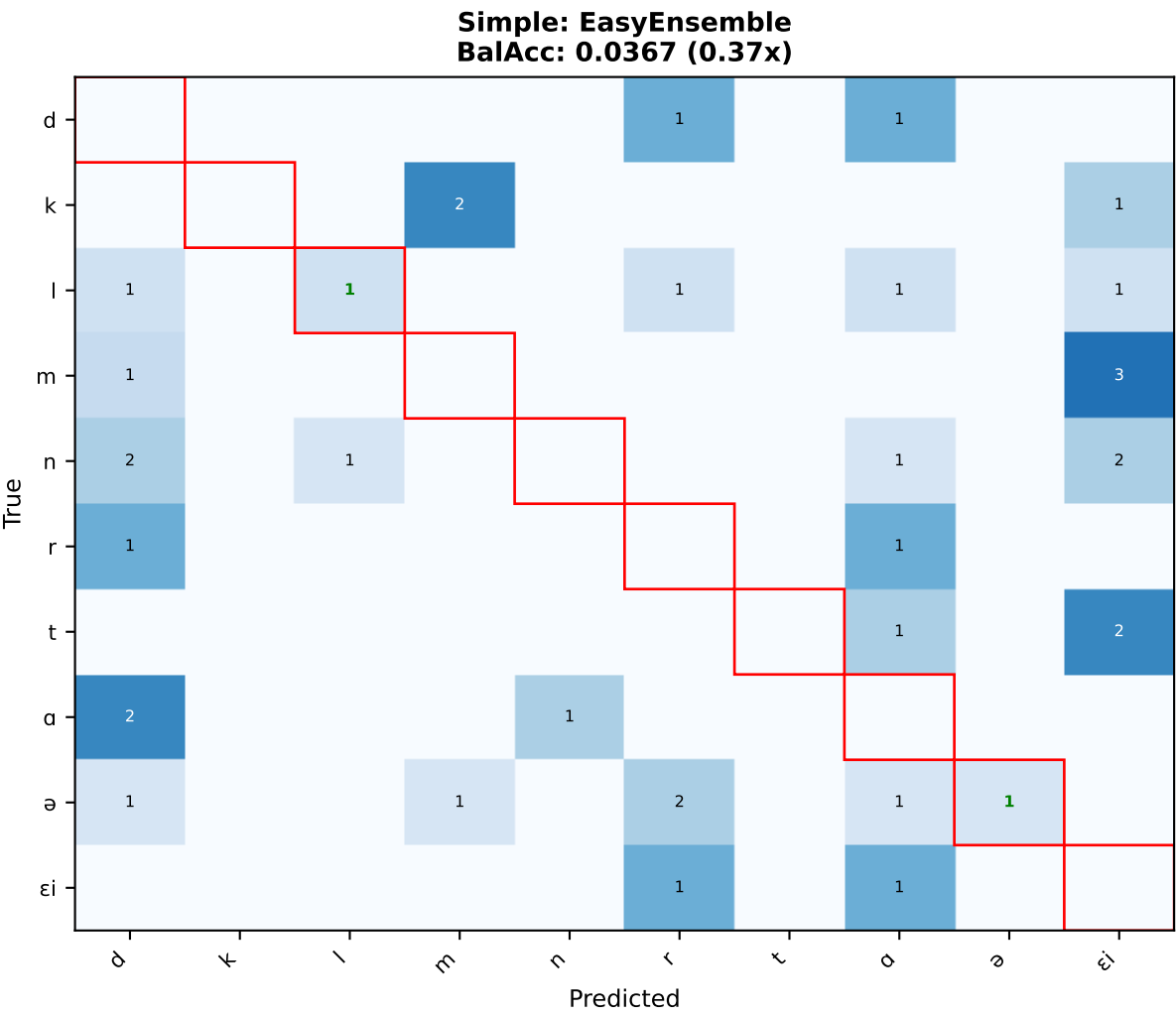


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	1	8	Fricative	v
Group 2	1	21	Short Vowel	ʌ
Group 3	2	42	Short Vowel	x, ə
Group 4	3	85	Plosive	d, n, t
Group 5	4	65	Liquid	l, r, s, ɹ

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	3	32	s, v, x
Liquid	2	42	l, r
Nasal	1	31	n
Plosive	2	54	d, t
Short Vowel	3	62	ʌ, ə, ɪ

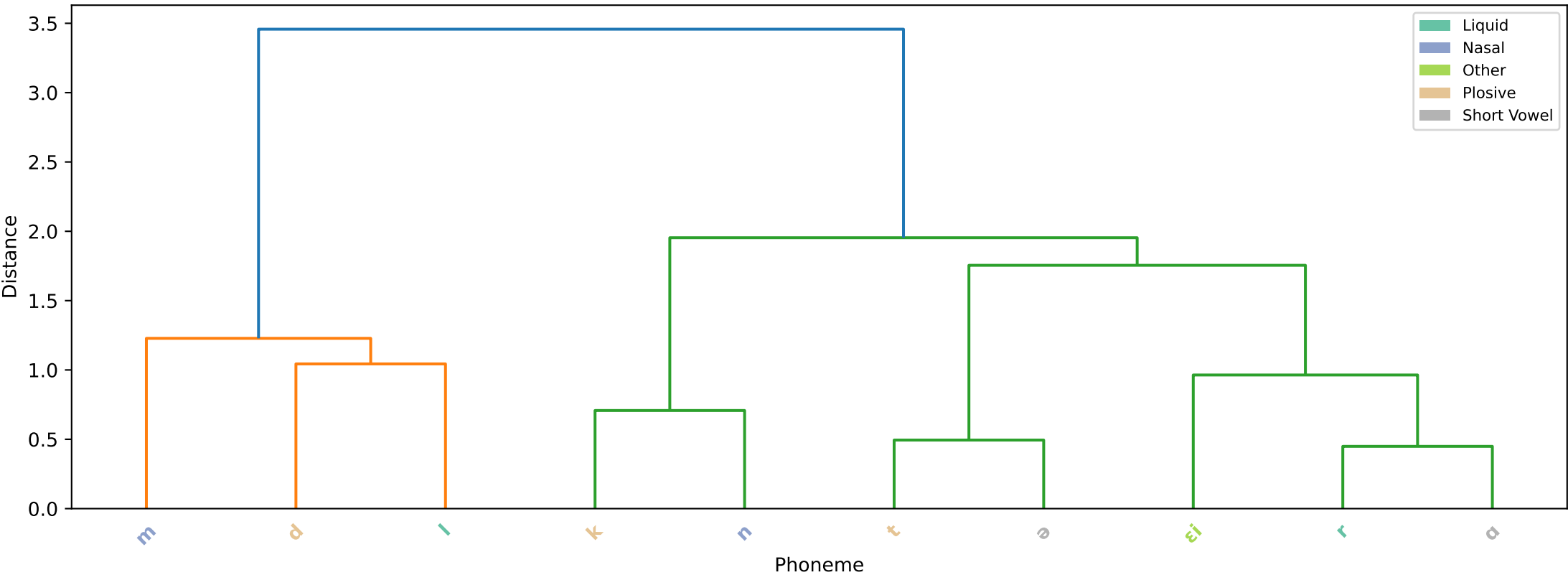


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
d	2	0.00	0.00	0.00	0.00	0.00	0.00	-
k	3	0.00	0.00	0.00	0.33	0.33	0.33	M
l	5	0.20	0.50	0.29	0.00	0.00	0.00	S
m	4	0.00	0.00	0.00	0.00	0.00	0.00	-
n	6	0.00	0.00	0.00	0.17	0.12	0.14	M
r	2	0.00	0.00	0.00	0.00	0.00	0.00	-
t	3	0.00	0.00	0.00	0.67	0.18	0.29	M
a	3	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	6	0.17	1.00	0.29	0.00	0.00	0.00	S
ei	2	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	36	0.04	0.15	0.06	0.12	0.06	0.08	M

Phoneme Clustering Analysis - P09

Hierarchical Clustering (Ward linkage)



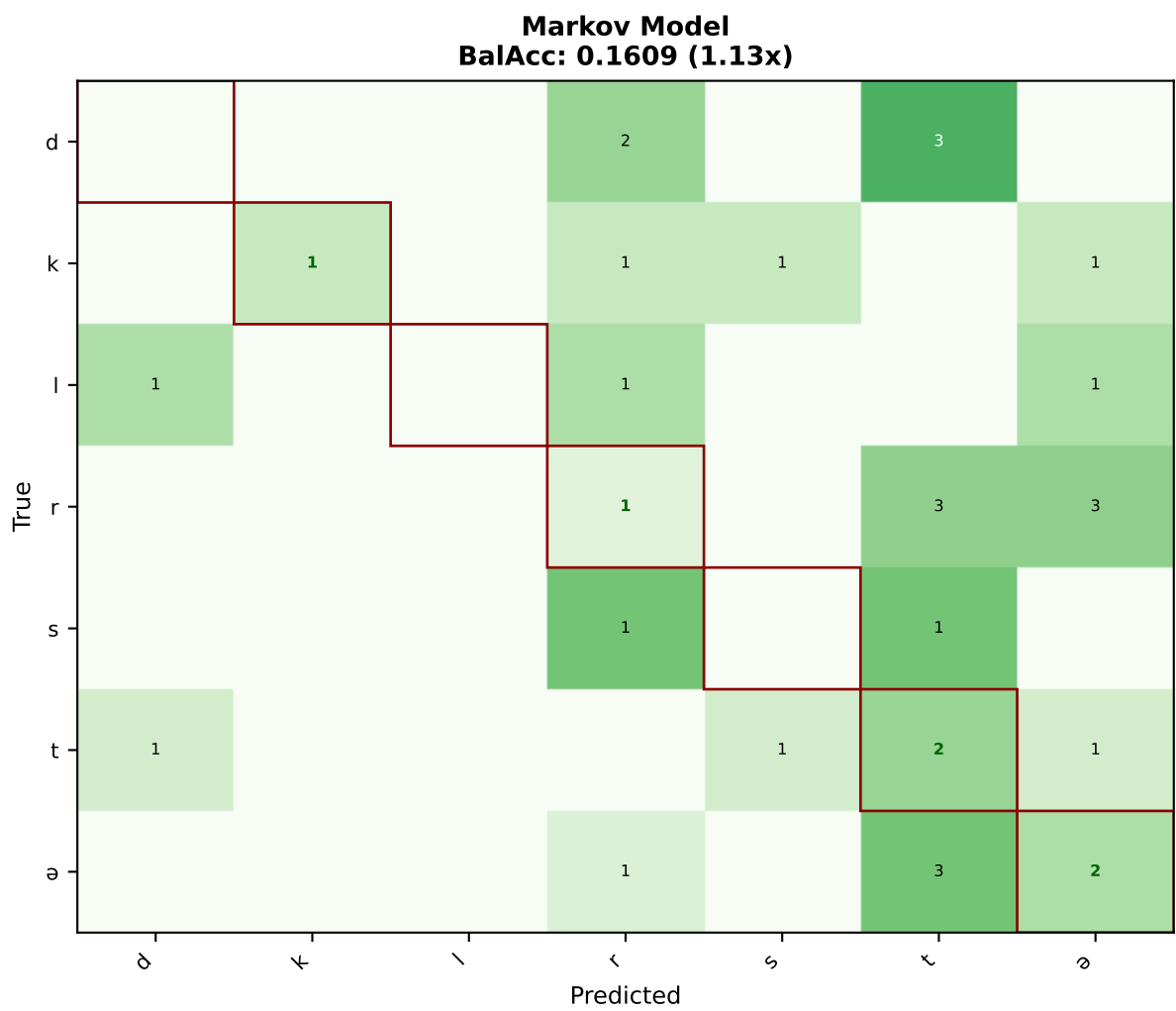
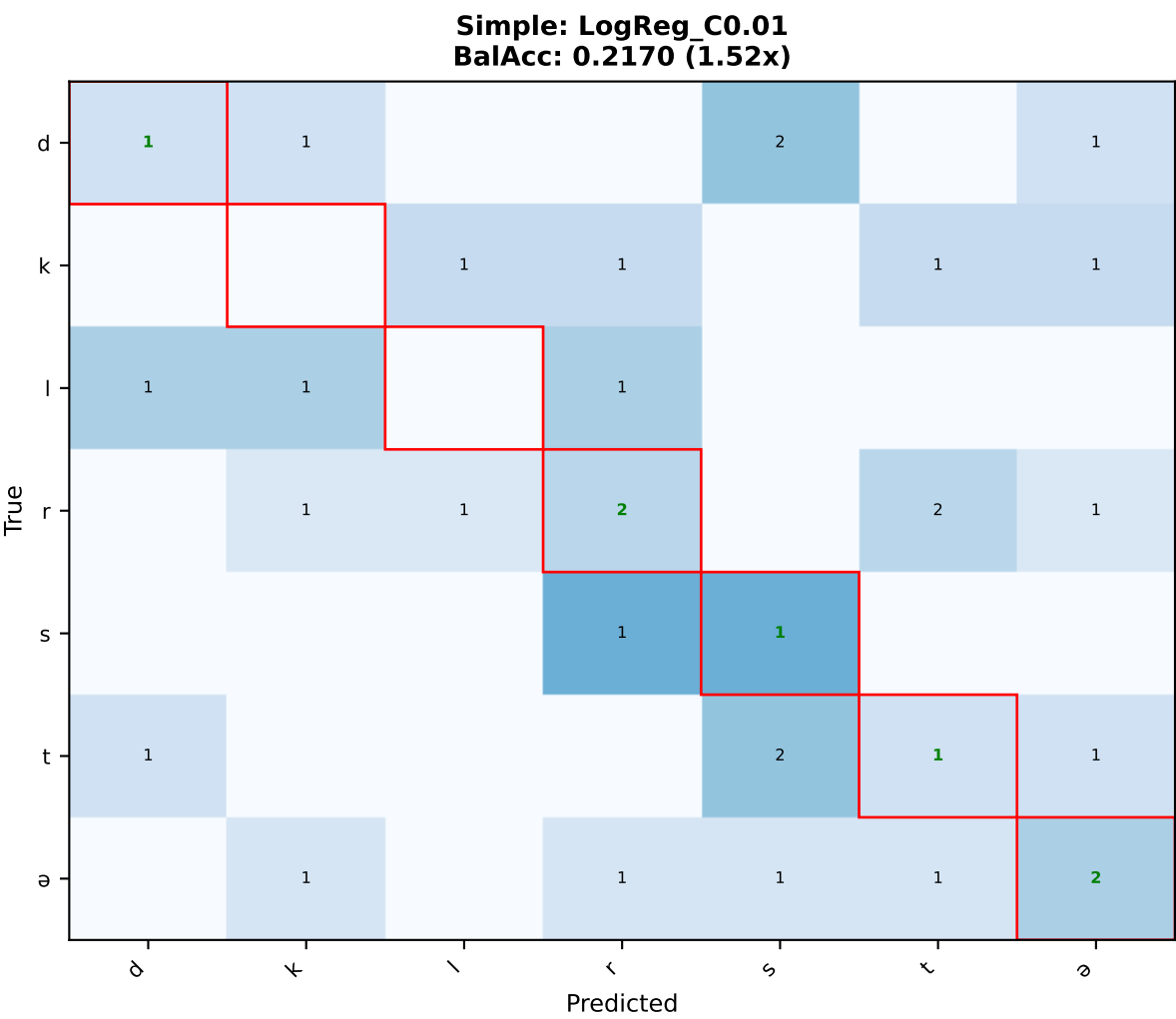
Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	2	32	Plosive	d, l
Group 2	1	11	Nasal	m
Group 3	2	50	Nasal	k, n
Group 4	2	72	Plosive	t, ə
Group 5	3	50	Liquid	r, ʌ, ei

Phonetic Categories

Category	N	Samples	Phonemes
Liquid	2	41	l, r
Nasal	2	44	m, n
Other	1	7	ei
Plosive	3	75	d, k, t
Short Vowel	2	48	ʌ, ə

Patient: P10 | Winner: Simple | Simple: 0.217 vs Markov: 0.161

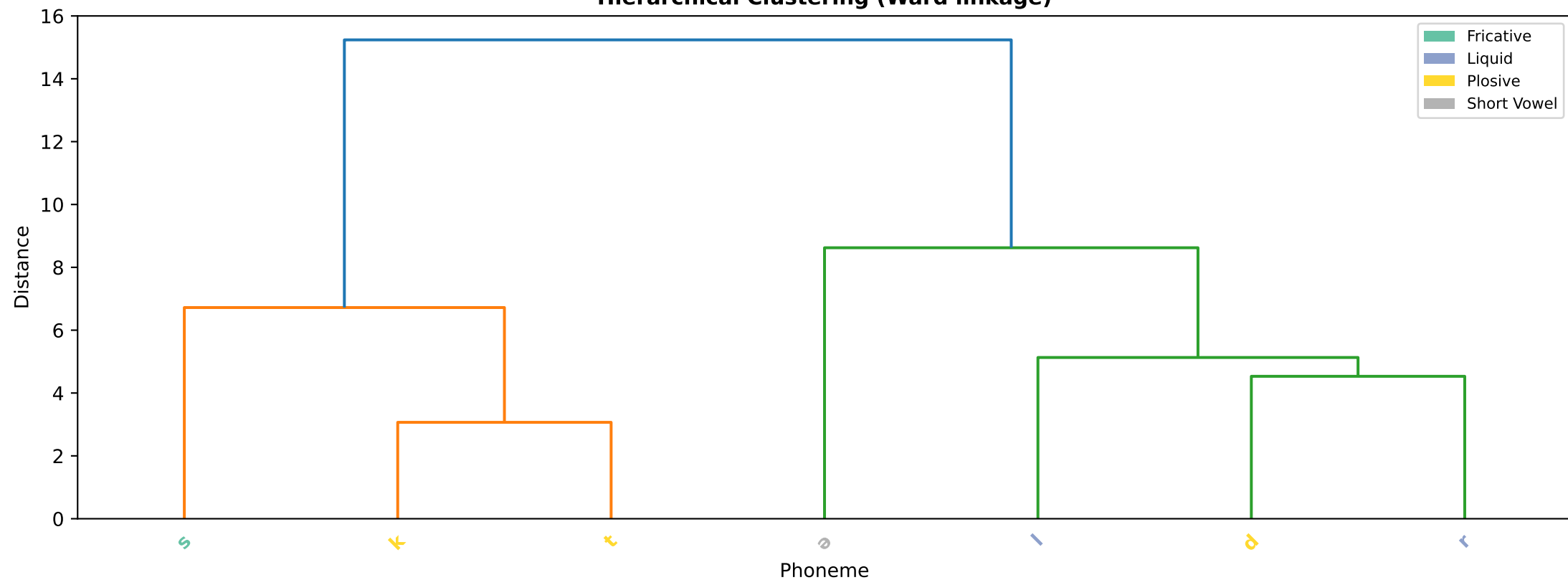


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
d	5	0.20	0.33	0.25	0.00	0.00	0.00	S
k	4	0.00	0.00	0.00	0.25	1.00	0.40	M
l	3	0.00	0.00	0.00	0.00	0.00	0.00	-
r	7	0.29	0.33	0.31	0.14	0.14	0.14	S
s	2	0.50	0.17	0.25	0.00	0.00	0.00	S
t	5	0.20	0.20	0.20	0.40	0.17	0.24	M
ə	6	0.33	0.33	0.33	0.33	0.25	0.29	S
TOTAL	32	0.22	0.20	0.19	0.16	0.22	0.15	S

Phoneme Clustering Analysis - P10

Hierarchical Clustering (Ward linkage)



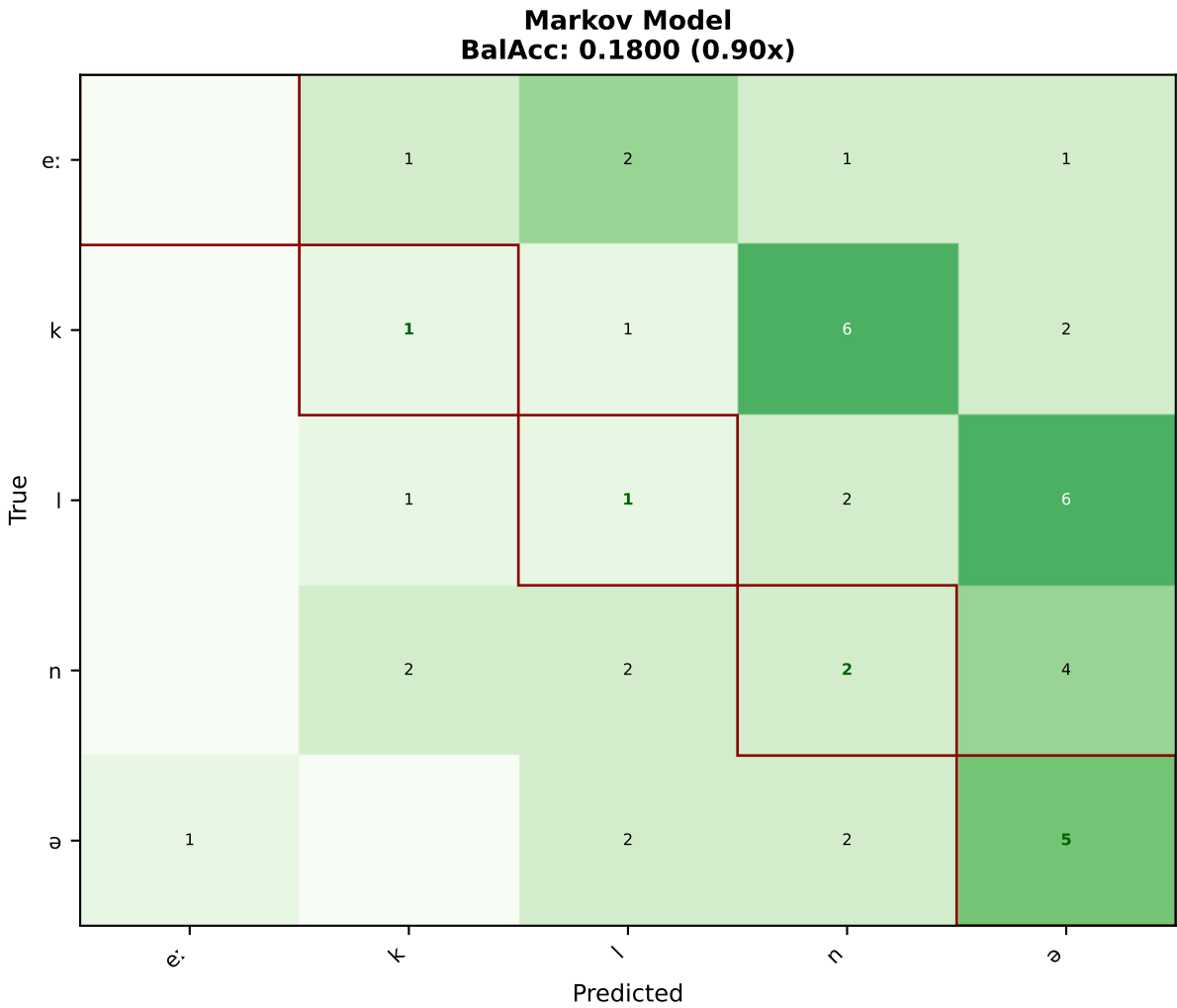
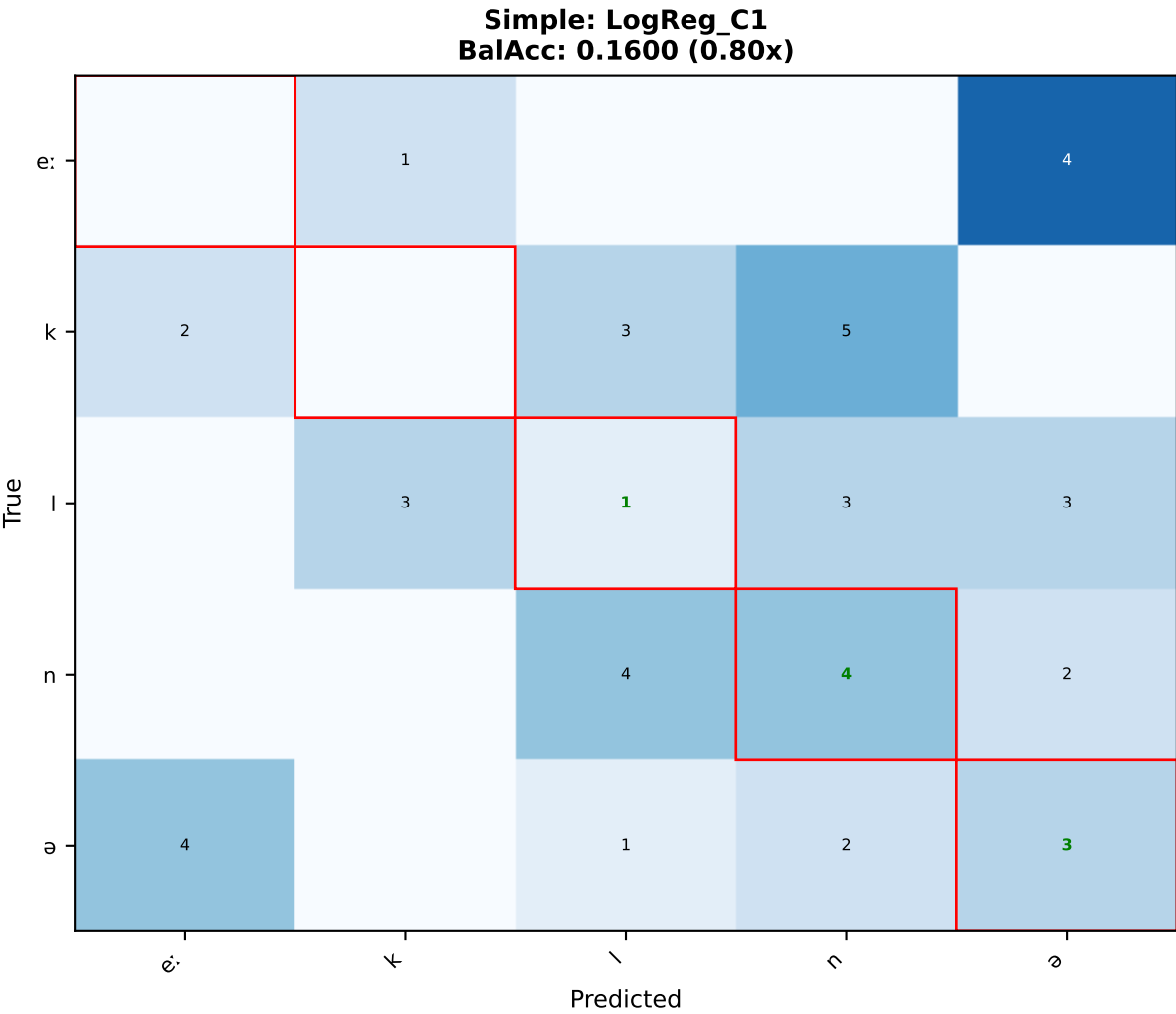
Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	2	43	Plosive	k, t
Group 2	1	15	Fricative	s
Group 3	2	37	Plosive	d, r
Group 4	1	17	Liquid	l
Group 5	1	31	Short Vowel	ə

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	1	15	s
Liquid	2	37	l, r
Plosive	3	60	d, k, t
Short Vowel	1	31	ə

Patient: P17 | Winner: Markov | Simple: 0.160 vs Markov: 0.180

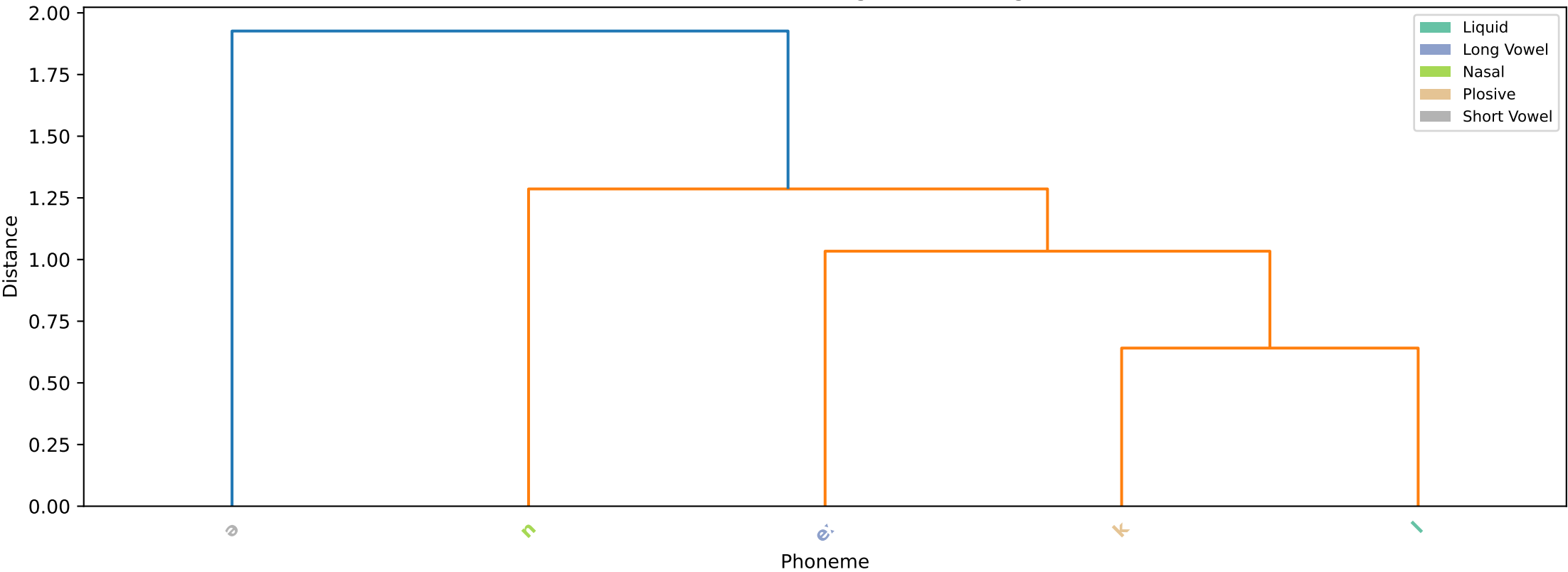


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
e:	5	0.00	0.00	0.00	0.00	0.00	0.00	-
k	10	0.00	0.00	0.00	0.10	0.20	0.13	M
l	10	0.10	0.11	0.11	0.10	0.12	0.11	M
n	10	0.40	0.29	0.33	0.20	0.15	0.17	S
ə	10	0.30	0.25	0.27	0.50	0.28	0.36	M
TOTAL	45	0.16	0.13	0.14	0.18	0.15	0.16	M

Phoneme Clustering Analysis - P17

Hierarchical Clustering (Ward linkage)

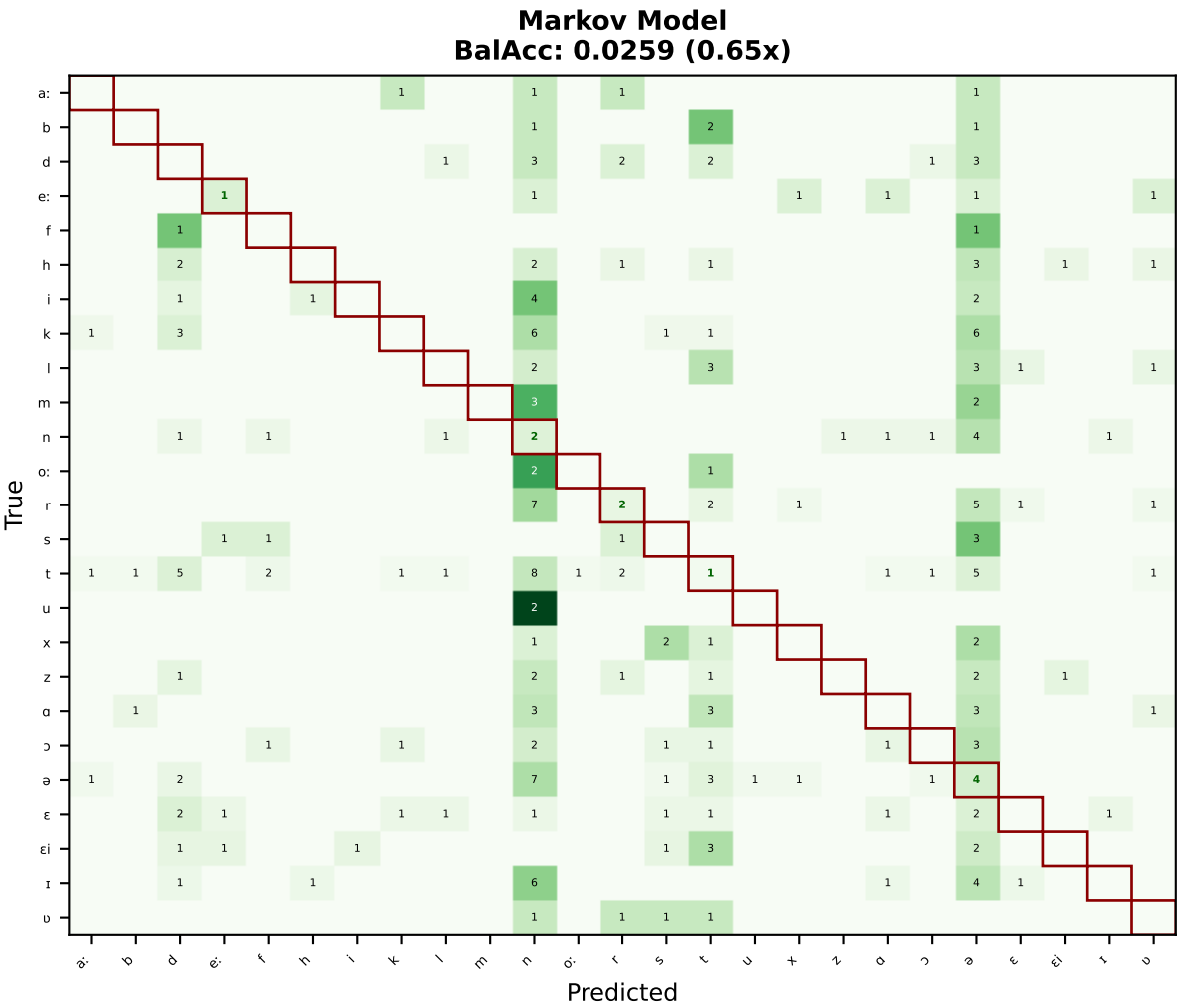
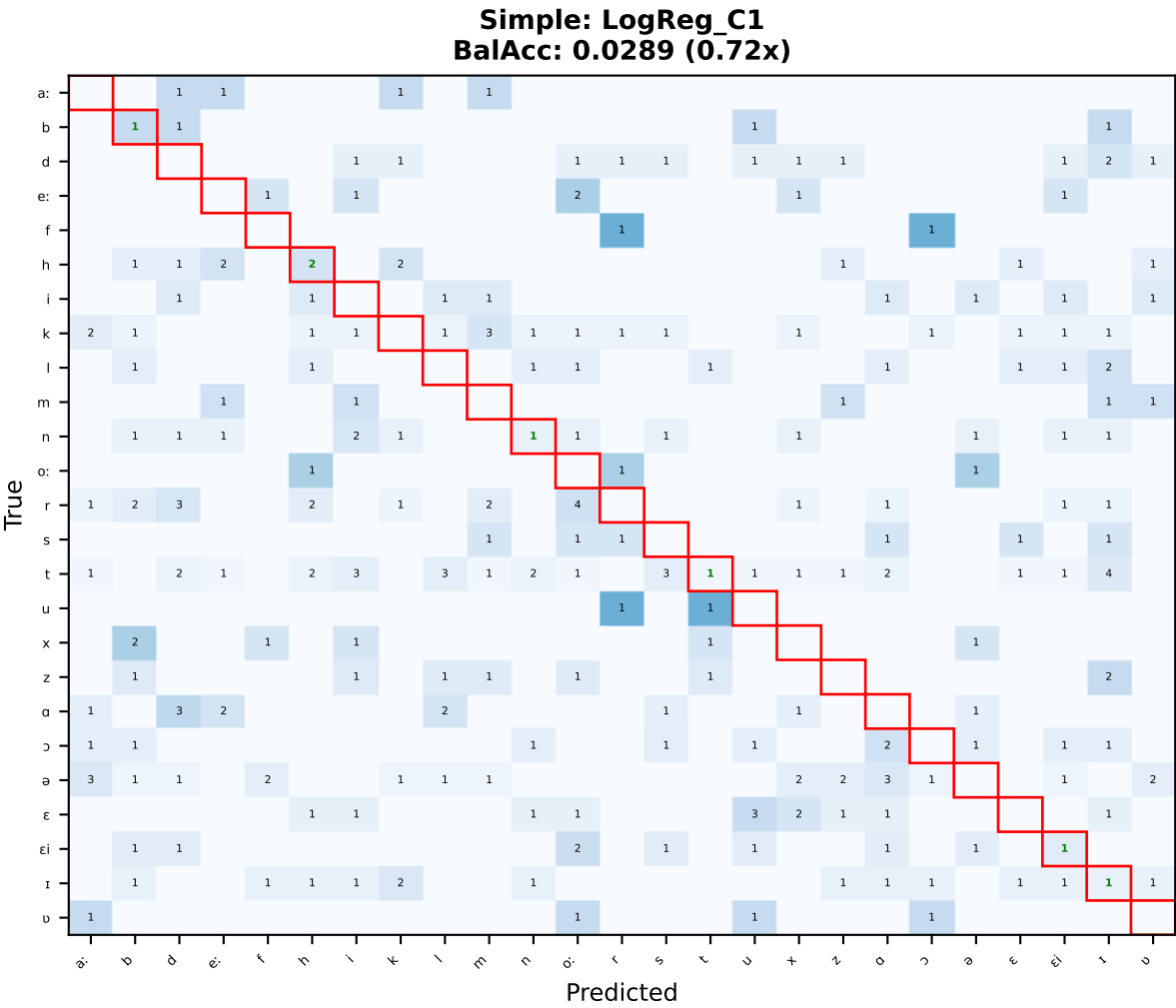


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	1	20	Long Vowel	e:
Group 2	1	20	Plosive	k
Group 3	1	20	Liquid	l
Group 4	1	35	Nasal	n
Group 5	1	25	Short Vowel	ə

Phonetic Categories

Category	N	Samples	Phonemes
Liquid	1	20	l
Long Vowel	1	20	e:
Nasal	1	35	n
Plosive	1	20	k
Short Vowel	1	25	ə

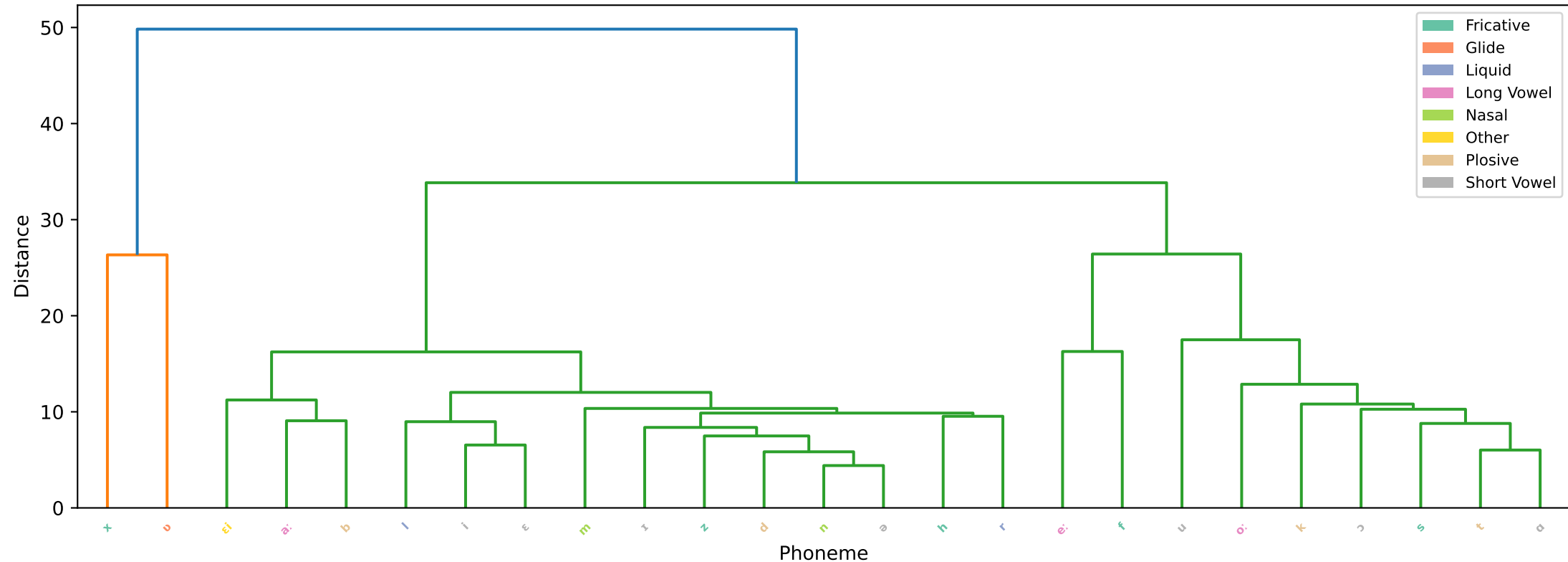


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
a:	4	0.00	0.00	0.00	0.00	0.00	0.00	-
b	4	0.25	0.07	0.11	0.00	0.00	0.00	S
d	12	0.00	0.00	0.00	0.00	0.00	0.00	-
e:	6	0.00	0.00	0.00	0.17	0.25	0.20	M
f	2	0.00	0.00	0.00	0.00	0.00	0.00	-
h	11	0.18	0.17	0.17	0.00	0.00	0.00	S
i	8	0.00	0.00	0.00	0.00	0.00	0.00	-
k	18	0.00	0.00	0.00	0.00	0.00	0.00	-
l	10	0.00	0.00	0.00	0.00	0.00	0.00	-
m	5	0.00	0.00	0.00	0.00	0.00	0.00	-
n	13	0.08	0.12	0.10	0.15	0.03	0.05	S
o:	3	0.00	0.00	0.00	0.00	0.00	0.00	-
r	19	0.00	0.00	0.00	0.11	0.18	0.13	M
s	6	0.00	0.00	0.00	0.00	0.00	0.00	-
t	31	0.03	0.20	0.06	0.03	0.04	0.03	S
u	2	0.00	0.00	0.00	0.00	0.00	0.00	-
x	6	0.00	0.00	0.00	0.00	0.00	0.00	-
z	8	0.00	0.00	0.00	0.00	0.00	0.00	-
ɑ	11	0.00	0.00	0.00	0.00	0.00	0.00	-
ɔ	10	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	21	0.00	0.00	0.00	0.19	0.06	0.10	M
ɛ	12	0.00	0.00	0.00	0.00	0.00	0.00	-
ɛɪ	9	0.11	0.08	0.10	0.00	0.00	0.00	S
ɪ	14	0.07	0.05	0.06	0.00	0.00	0.00	S
ʊ	4	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	249	0.03	0.03	0.02	0.03	0.02	0.02	S

Phoneme Clustering Analysis - P21

Hierarchical Clustering (Ward linkage)

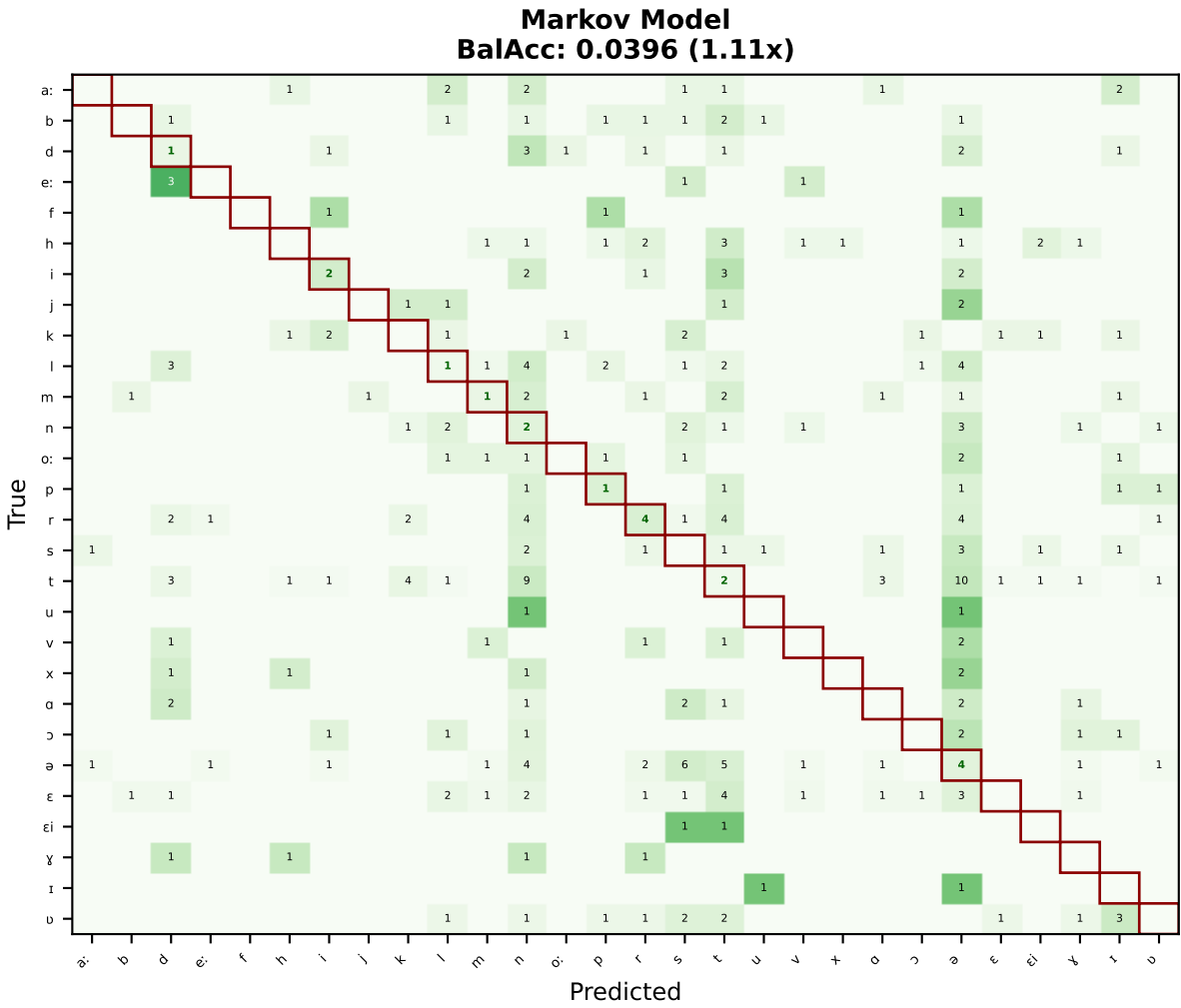
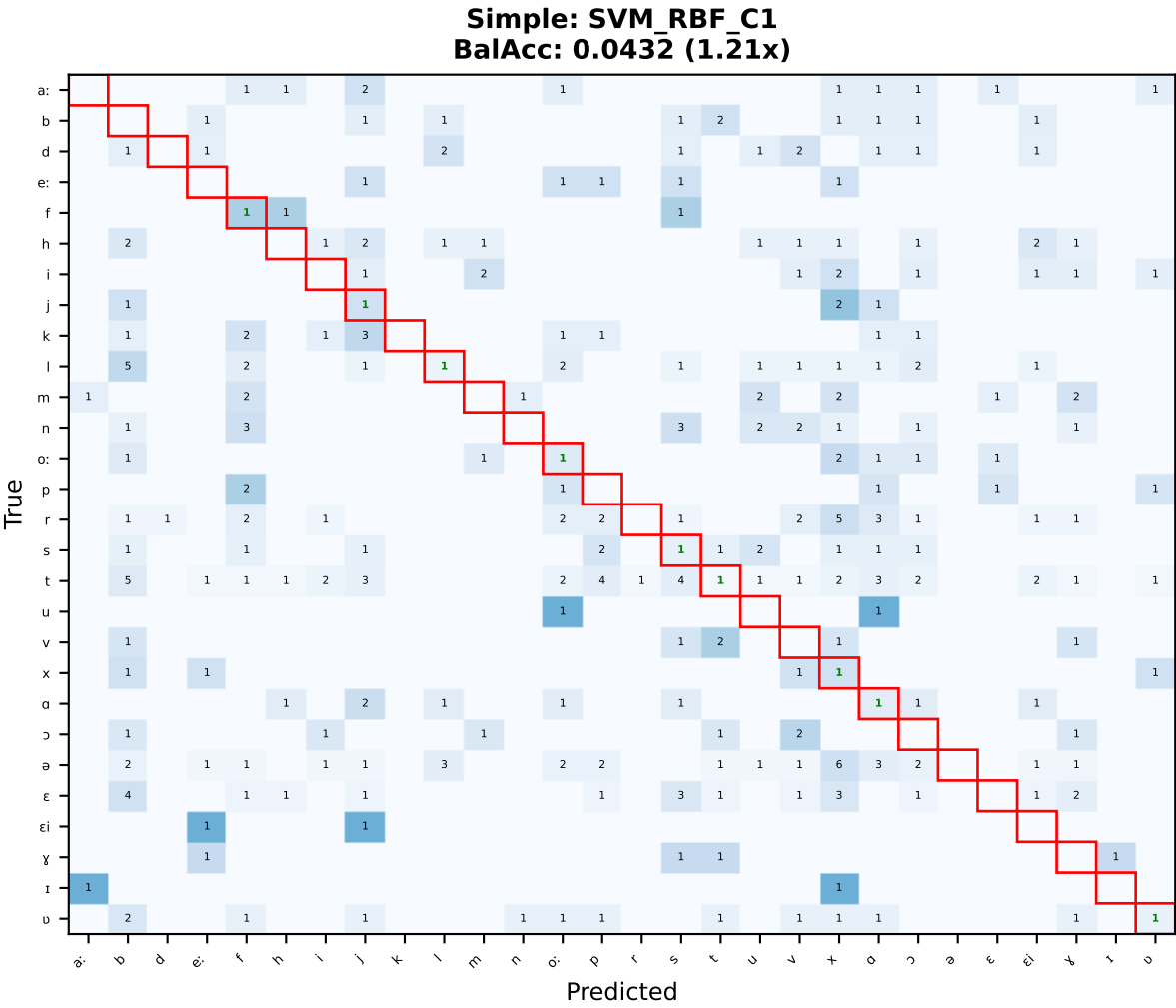


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	1	18	Fricative	x
Group 2	1	26	Glide	ʊ
Group 3	14	956	Short Vowel	a:, b, d, h, i, l, m, n, ...
Group 4	2	56	Long Vowel	e:, f
Group 5	7	394	Short Vowel	k, o:, s, t, u, a, ɔ

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	5	187	f, h, s, x, z
Glide	1	26	ʊ
Liquid	2	163	l, r
Long Vowel	3	93	a:, e:, ɔ:
Nasal	2	210	m, n
Other	1	33	ɛi
Plosive	4	299	b, d, k, t
Short Vowel	7	439	i, u, ɔ, ɔ, ə, ɛ, ɪ

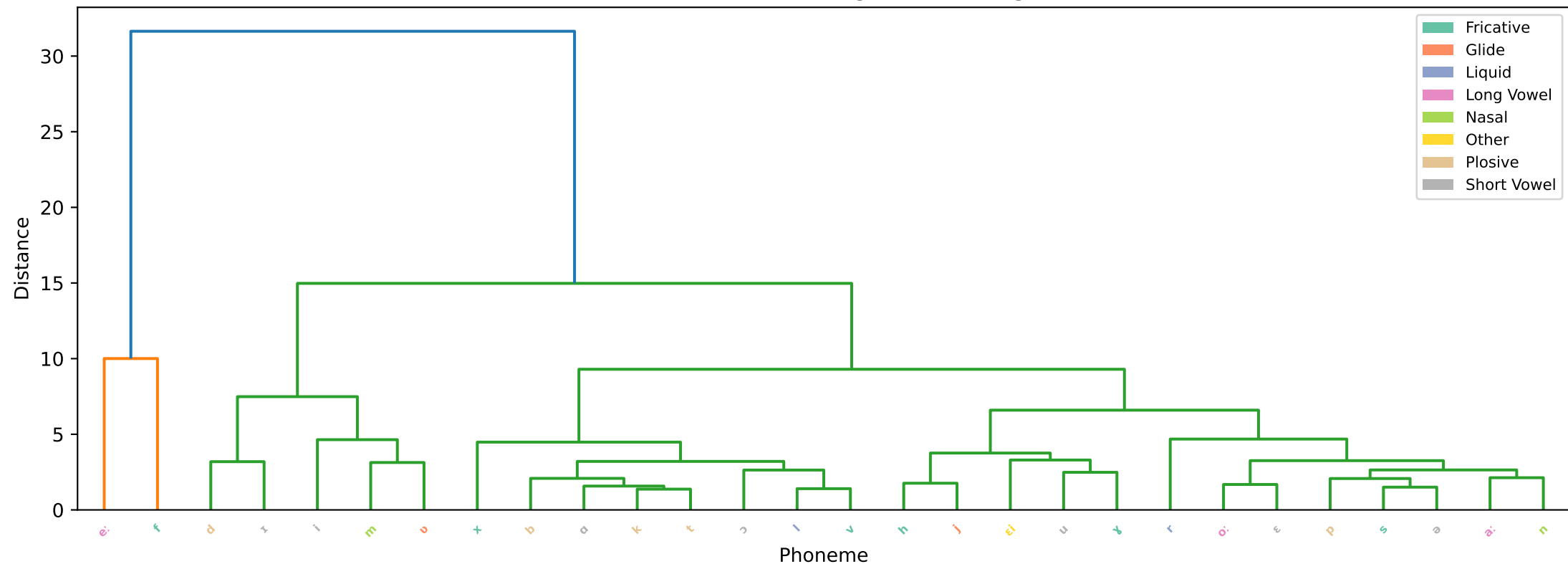


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
a:	10	0.00	0.00	0.00	0.00	0.00	0.00	-
b	10	0.00	0.00	0.00	0.00	0.00	0.00	-
d	11	0.00	0.00	0.00	0.09	0.05	0.07	M
e:	5	0.00	0.00	0.00	0.00	0.00	0.00	-
f	3	0.33	0.05	0.09	0.00	0.00	0.00	S
h	14	0.00	0.00	0.00	0.00	0.00	0.00	-
i	10	0.00	0.00	0.00	0.20	0.22	0.21	M
j	5	0.20	0.05	0.07	0.00	0.00	0.00	S
k	11	0.00	0.00	0.00	0.00	0.00	0.00	-
l	19	0.05	0.11	0.07	0.05	0.07	0.06	S
m	11	0.00	0.00	0.00	0.09	0.14	0.11	M
n	14	0.00	0.00	0.00	0.14	0.04	0.07	M
o:	8	0.12	0.06	0.08	0.00	0.00	0.00	S
p	6	0.00	0.00	0.00	0.17	0.12	0.14	M
r	23	0.00	0.00	0.00	0.17	0.24	0.20	M
s	12	0.08	0.05	0.06	0.00	0.00	0.00	S
t	38	0.03	0.09	0.04	0.05	0.05	0.05	M
u	2	0.00	0.00	0.00	0.00	0.00	0.00	-
v	6	0.00	0.00	0.00	0.00	0.00	0.00	-
x	5	0.20	0.03	0.05	0.00	0.00	0.00	S
ɑ	9	0.11	0.05	0.07	0.00	0.00	0.00	S
ɔ	7	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	29	0.00	0.00	0.00	0.14	0.07	0.10	M
ɛ	20	0.00	0.00	0.00	0.00	0.00	0.00	-
ɛɪ	2	0.00	0.00	0.00	0.00	0.00	0.00	-
ʏ	4	0.00	0.00	0.00	0.00	0.00	0.00	-
ɪ	2	0.00	0.00	0.00	0.00	0.00	0.00	-
ʊ	13	0.08	0.17	0.11	0.00	0.00	0.00	S
TOTAL	309	0.04	0.02	0.02	0.04	0.04	0.04	S

Phoneme Clustering Analysis - P22

Hierarchical Clustering (Ward linkage)

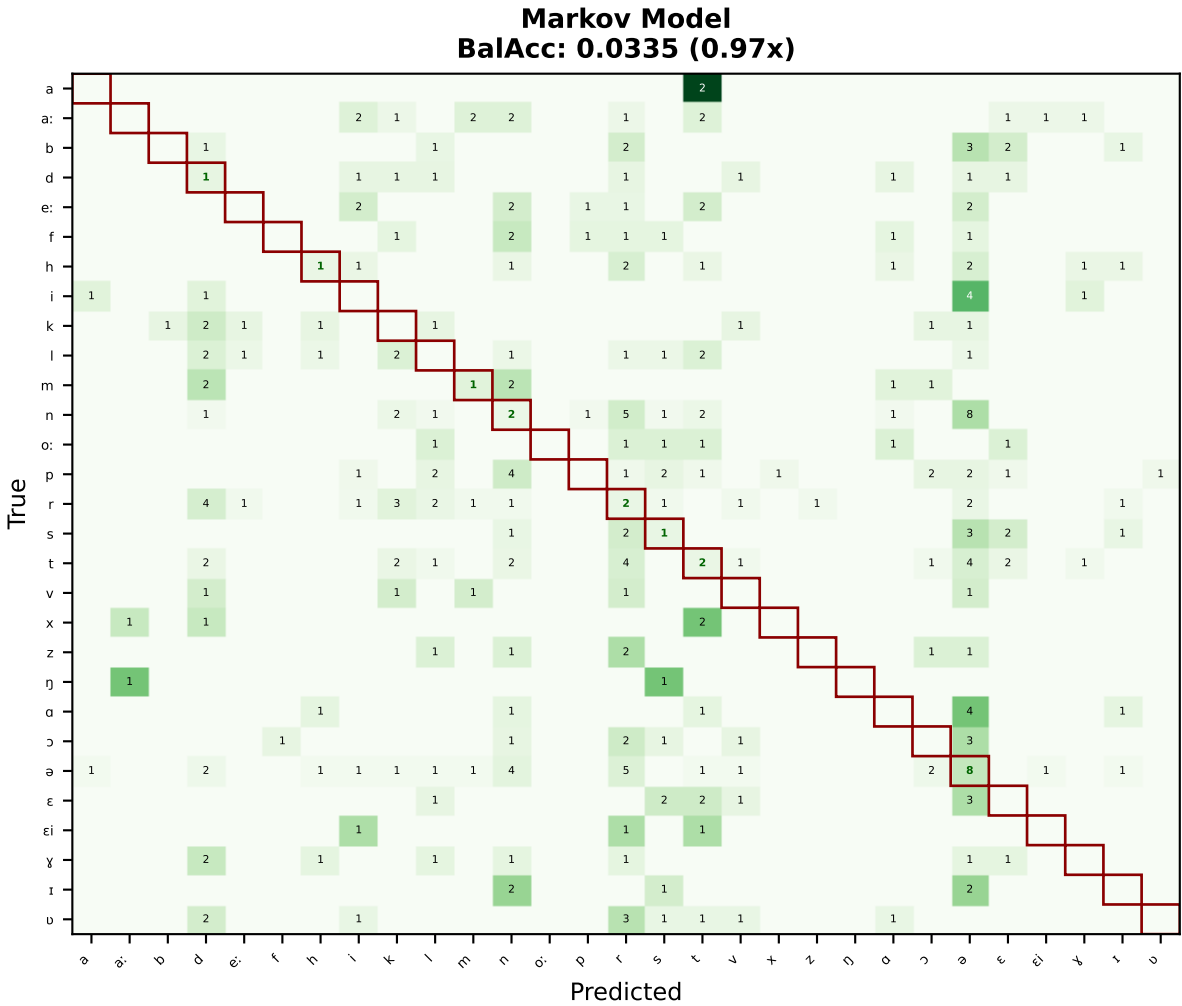
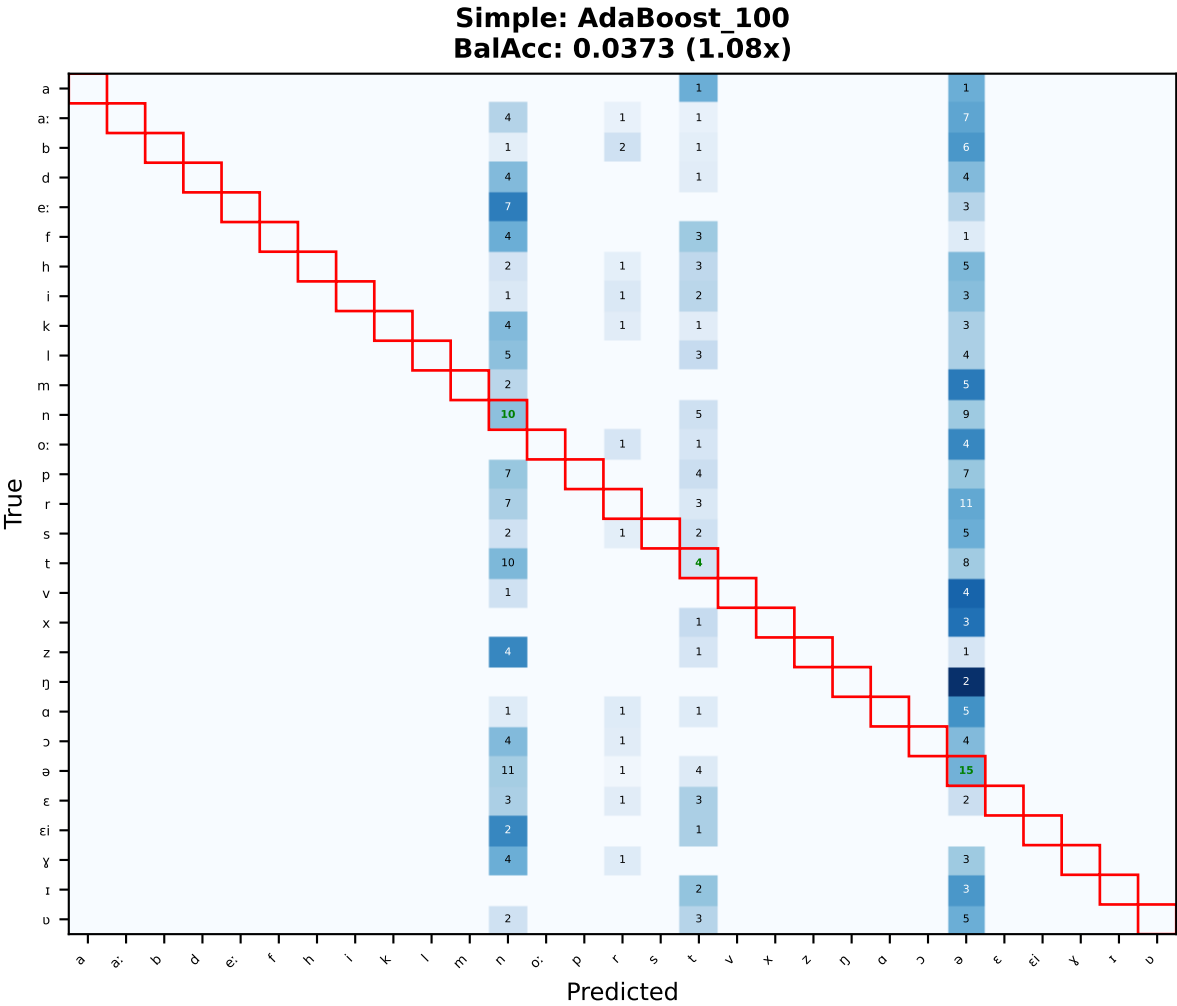


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	1	36	Long Vowel	e:
Group 2	1	22	Fricative	f
Group 3	5	305	Short Vowel	d, i, m, I, v
Group 4	8	563	Plosive	b, k, l, t, v, x, a, ɒ
Group 5	13	1014	Short Vowel	a:, h, j, n, o:, p, r, s, ...

Phonetic Categories

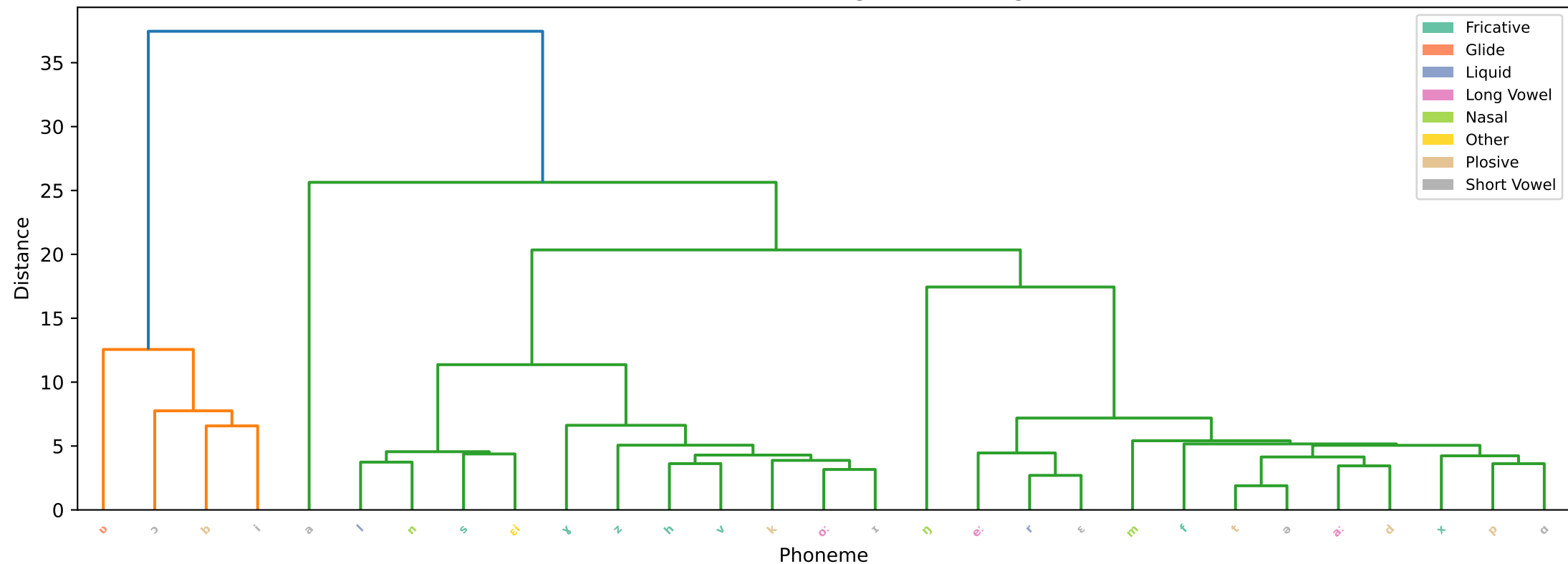
Category	N	Samples	Phonemes
Fricative	6	278	f, h, s, v, x, ʒ
Glide	2	45	j, v
Liquid	2	207	l, r
Long Vowel	3	115	a:, e:, o:
Nasal	2	256	m, n
Other	1	34	ɛi
Plosive	5	422	b, d, k, p, t
Short Vowel	7	583	i, u, a, ɔ, ə, ɛ, ɪ



Per-Phoneme Comparison (S=Simple, M=Markov)								
Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
a	2	0.00	0.00	0.00	0.00	0.00	0.00	-
a:	13	0.00	0.00	0.00	0.00	0.00	0.00	-
b	10	0.00	0.00	0.00	0.00	0.00	0.00	-
d	9	0.00	0.00	0.00	0.11	0.04	0.06	M
e:	10	0.00	0.00	0.00	0.00	0.00	0.00	-
f	8	0.00	0.00	0.00	0.00	0.00	0.00	-
h	11	0.00	0.00	0.00	0.09	0.17	0.12	M
i	7	0.00	0.00	0.00	0.00	0.00	0.00	-
k	9	0.00	0.00	0.00	0.00	0.00	0.00	-
l	12	0.00	0.00	0.00	0.00	0.00	0.00	-
m	7	0.00	0.00	0.00	0.14	0.17	0.15	M
n	24	0.42	0.10	0.16	0.08	0.07	0.07	S
o:	6	0.00	0.00	0.00	0.00	0.00	0.00	-
p	18	0.00	0.00	0.00	0.00	0.00	0.00	-
r	21	0.00	0.00	0.00	0.10	0.05	0.07	M
s	10	0.00	0.00	0.00	0.10	0.07	0.08	M
t	22	0.18	0.08	0.11	0.09	0.09	0.09	S
v	5	0.00	0.00	0.00	0.00	0.00	0.00	-
x	4	0.00	0.00	0.00	0.00	0.00	0.00	-
z	6	0.00	0.00	0.00	0.00	0.00	0.00	-
ʒ	2	0.00	0.00	0.00	0.00	0.00	0.00	-
ɑ	8	0.00	0.00	0.00	0.00	0.00	0.00	-
ɔ	9	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	31	0.48	0.11	0.18	0.26	0.14	0.18	S
ɛ	9	0.00	0.00	0.00	0.00	0.00	0.00	-
ɛi	3	0.00	0.00	0.00	0.00	0.00	0.00	-
ʏ	8	0.00	0.00	0.00	0.00	0.00	0.00	-
ɪ	5	0.00	0.00	0.00	0.00	0.00	0.00	-
ʊ	10	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	299	0.04	0.01	0.02	0.03	0.03	0.03	S

Phoneme Clustering Analysis - P24

Hierarchical Clustering (Ward linkage)



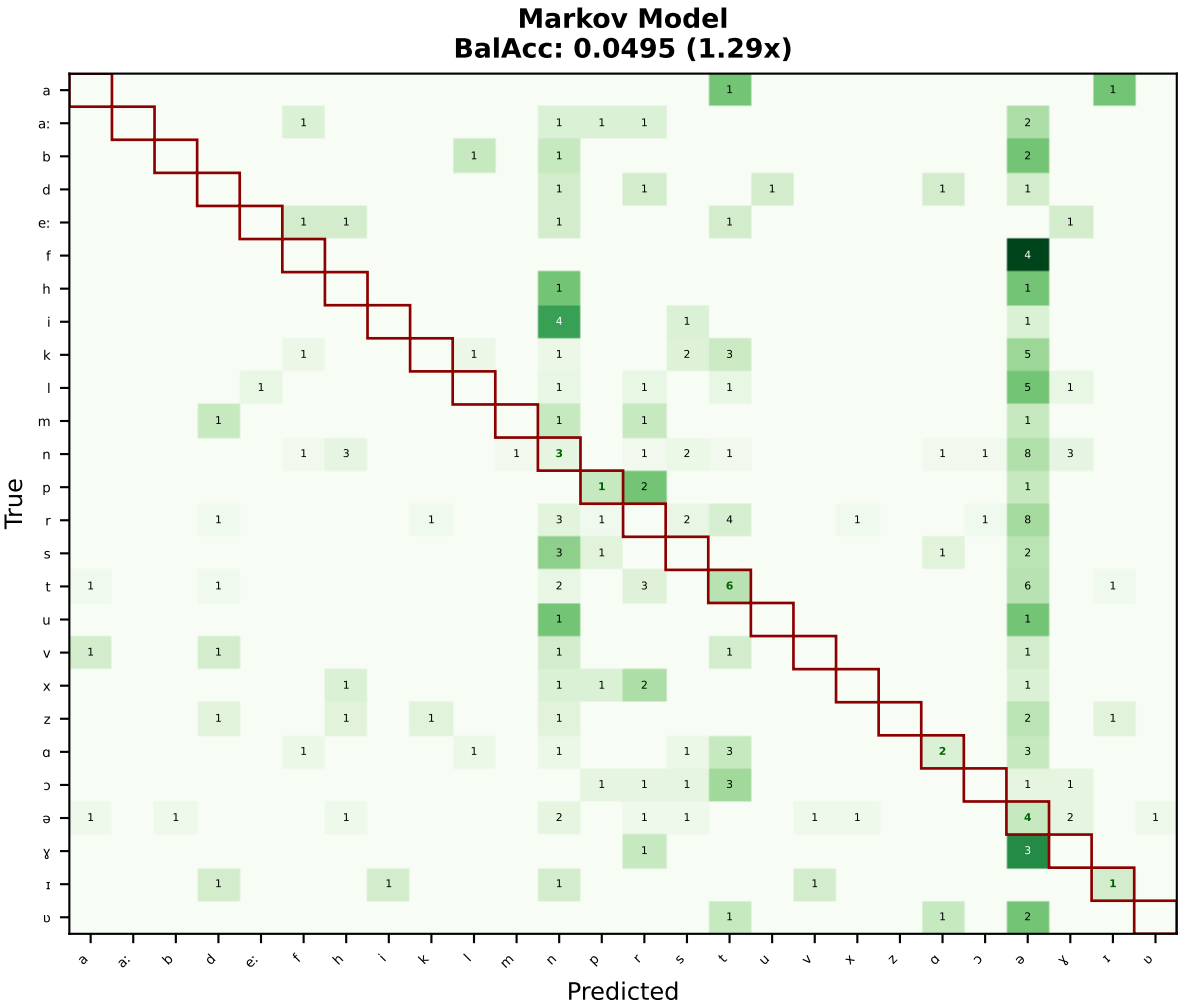
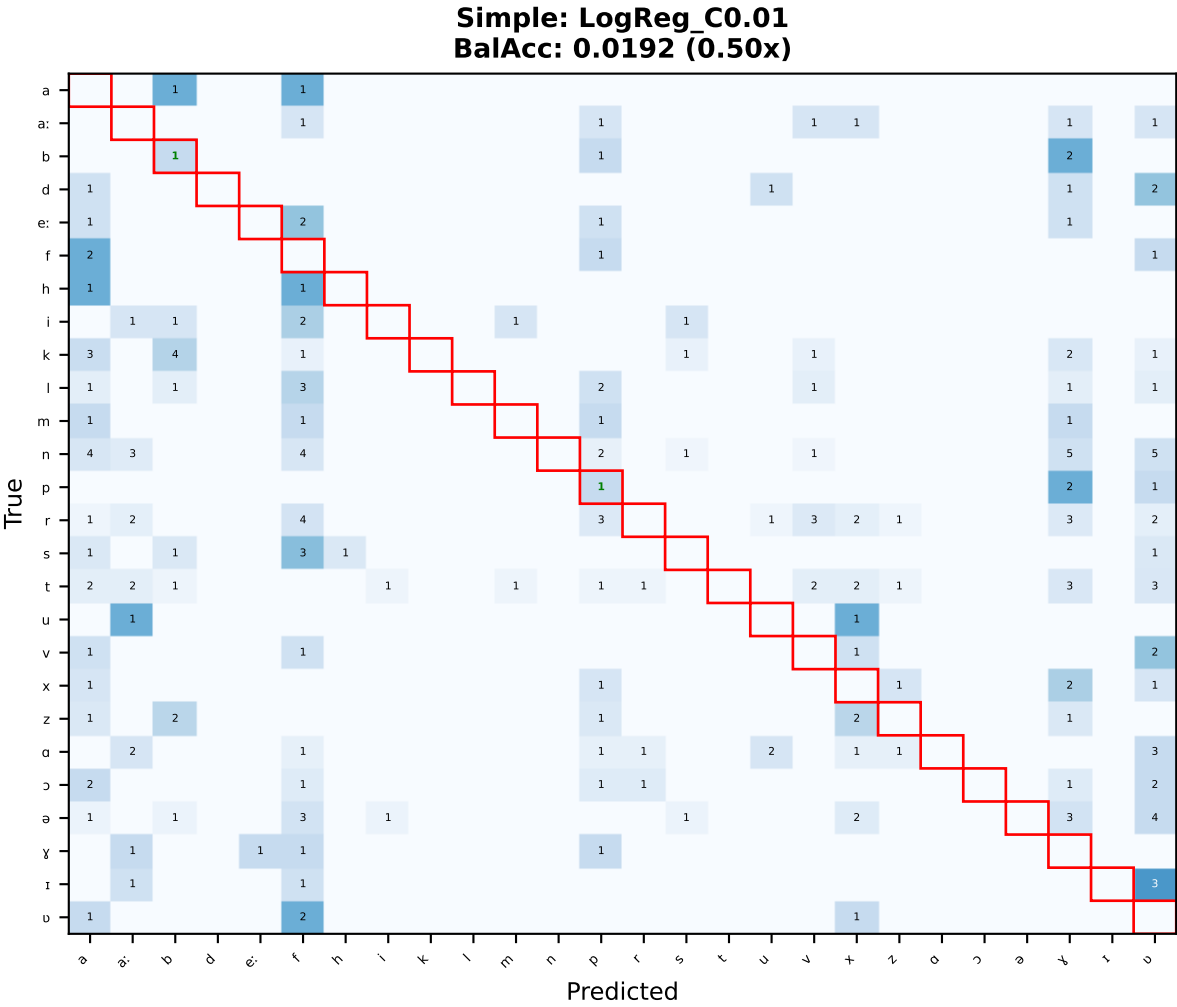
Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	4	149	Short Vowel	b, i, ɔ, ʊ
Group 2	11	703	Fricative	h, k, l, n, o:, s, v, z, ...
Group 3	12	979	Short Vowel	a:, d, e:, f, m, p, r, t,...
Group 4	1	13	Nasal	ŋ
Group 5	1	7	Short Vowel	a

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	7	293	f, h, s, v, x, z, ʒ
Glide	1	31	ʊ
Liquid	2	201	l, r
Long Vowel	3	103	a:, e:, o:
Nasal	3	244	m, n, ŋ
Other	1	32	ɛi
Plosive	5	405	b, d, k, p, t
Short Vowel	7	542	a, i, ɔ, ɔ, ə, ɛ, ɪ

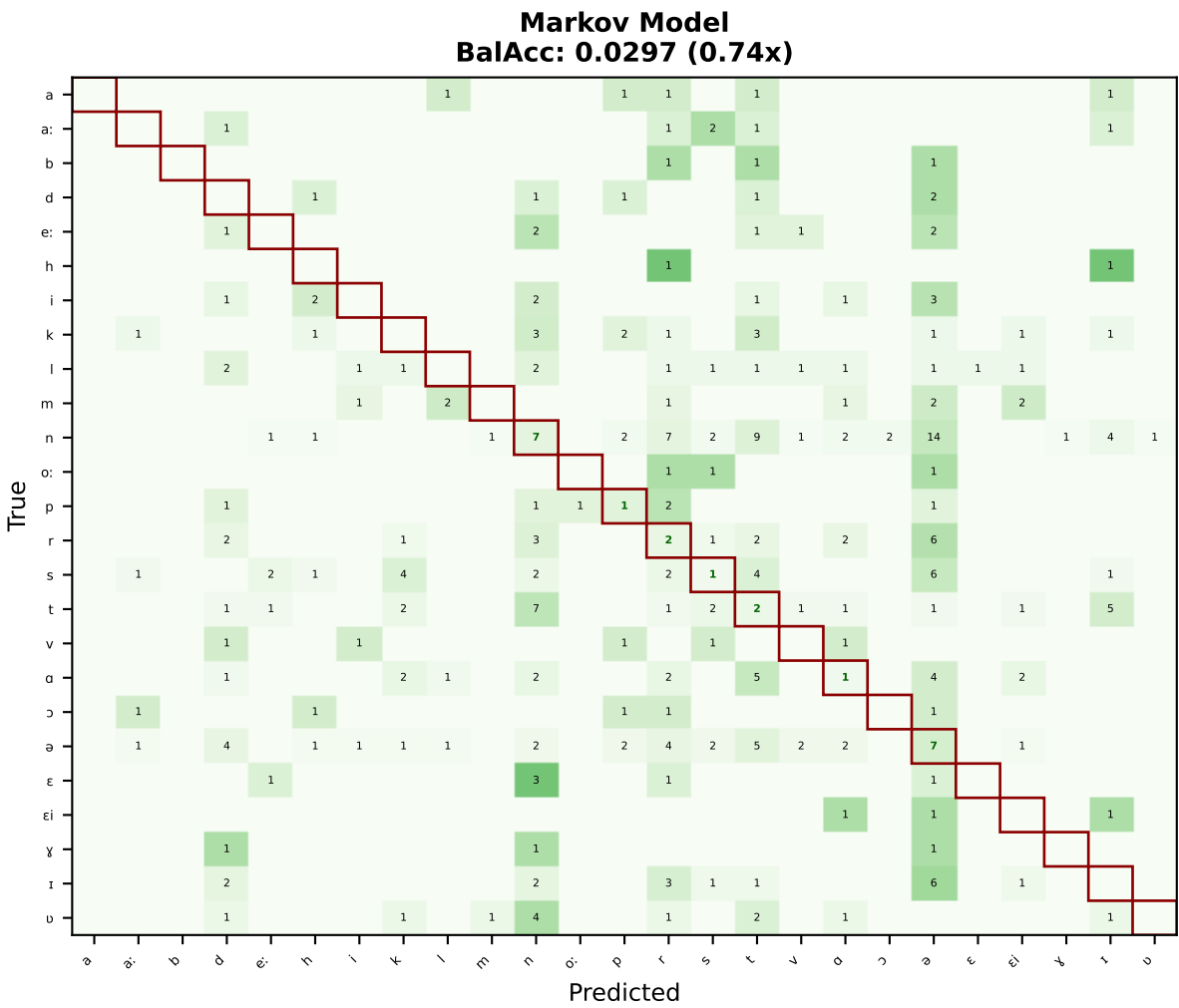
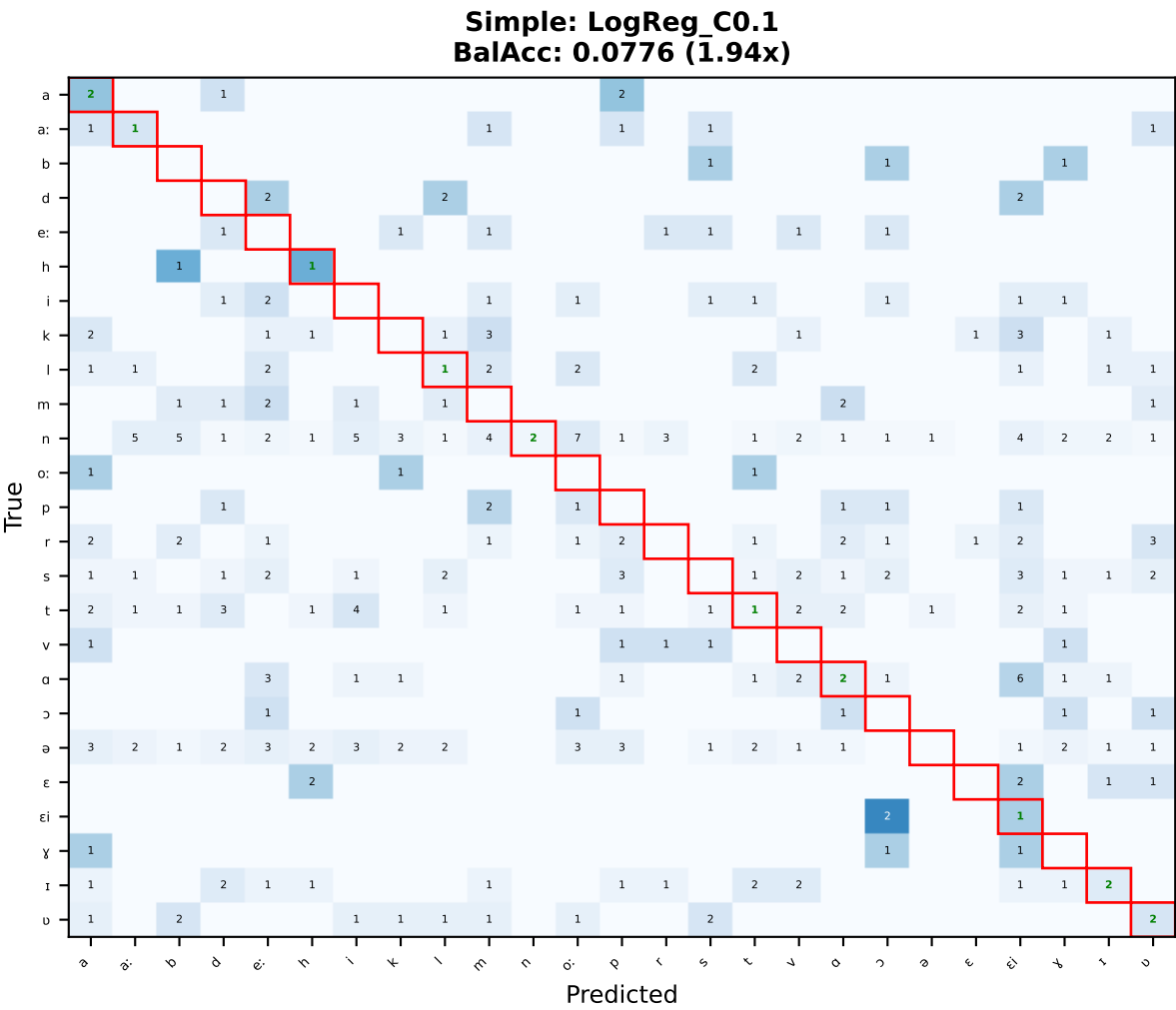
Patient: P26 | Winner: Markov | Simple: 0.019 vs Markov: 0.049



Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
a	2	0.00	0.00	0.00	0.00	0.00	0.00	-
a:	6	0.00	0.00	0.00	0.00	0.00	0.00	-
b	4	0.25	0.08	0.12	0.00	0.00	0.00	S
d	5	0.00	0.00	0.00	0.00	0.00	0.00	-
e:	5	0.00	0.00	0.00	0.00	0.00	0.00	-
f	4	0.00	0.00	0.00	0.00	0.00	0.00	-
h	2	0.00	0.00	0.00	0.00	0.00	0.00	-
i	6	0.00	0.00	0.00	0.00	0.00	0.00	-
k	13	0.00	0.00	0.00	0.00	0.00	0.00	-
l	10	0.00	0.00	0.00	0.00	0.00	0.00	-
m	4	0.00	0.00	0.00	0.00	0.00	0.00	-
n	25	0.00	0.00	0.00	0.12	0.10	0.11	M
p	4	0.25	0.05	0.09	0.25	0.17	0.20	M
r	22	0.00	0.00	0.00	0.00	0.00	0.00	-
s	7	0.00	0.00	0.00	0.00	0.00	0.00	-
t	20	0.00	0.00	0.00	0.30	0.24	0.27	M
u	2	0.00	0.00	0.00	0.00	0.00	0.00	-
v	5	0.00	0.00	0.00	0.00	0.00	0.00	-
x	6	0.00	0.00	0.00	0.00	0.00	0.00	-
z	7	0.00	0.00	0.00	0.00	0.00	0.00	-
ɔ	12	0.00	0.00	0.00	0.17	0.33	0.22	M
ɔ:	8	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	16	0.00	0.00	0.00	0.25	0.06	0.10	M
ʏ	4	0.00	0.00	0.00	0.00	0.00	0.00	-
ɪ	5	0.00	0.00	0.00	0.20	0.25	0.22	M
ʊ	4	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	208	0.02	0.00	0.01	0.05	0.04	0.04	M

Patient: P28 | Winner: Simple | Simple: 0.078 vs Markov: 0.030

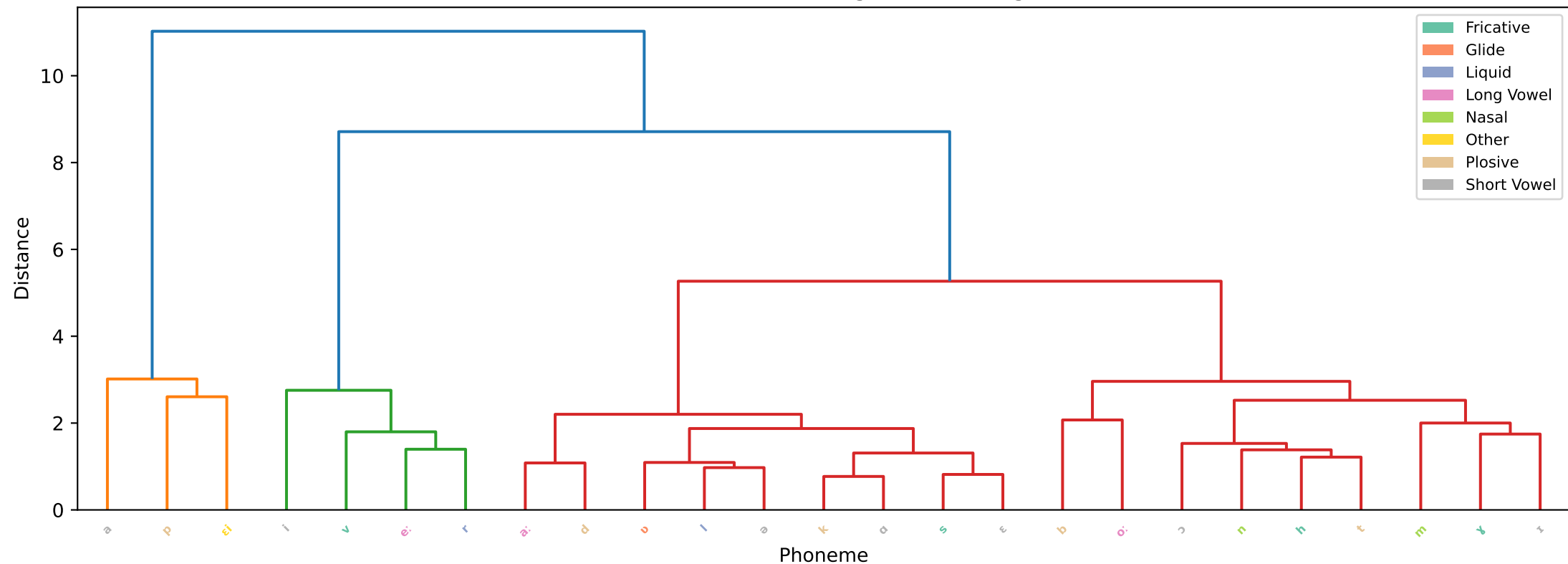


Per-Phoneme Comparison (S=Simple, M=Markov)

Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
a	5	0.40	0.11	0.17	0.00	0.00	0.00	S
a:	6	0.17	0.09	0.12	0.00	0.00	0.00	S
b	3	0.00	0.00	0.00	0.00	0.00	0.00	-
d	6	0.00	0.00	0.00	0.00	0.00	0.00	-
e:	7	0.00	0.00	0.00	0.00	0.00	0.00	-
h	2	0.50	0.11	0.18	0.00	0.00	0.00	S
i	10	0.00	0.00	0.00	0.00	0.00	0.00	-
k	14	0.00	0.00	0.00	0.00	0.00	0.00	-
l	14	0.07	0.08	0.08	0.00	0.00	0.00	S
m	9	0.00	0.00	0.00	0.00	0.00	0.00	-
n	55	0.04	1.00	0.07	0.13	0.16	0.14	M
o:	3	0.00	0.00	0.00	0.00	0.00	0.00	-
p	7	0.00	0.00	0.00	0.14	0.09	0.11	M
r	19	0.00	0.00	0.00	0.11	0.06	0.08	M
s	24	0.00	0.00	0.00	0.04	0.07	0.05	M
t	25	0.04	0.08	0.05	0.08	0.05	0.06	M
v	5	0.00	0.00	0.00	0.00	0.00	0.00	-
ɔ	20	0.10	0.15	0.12	0.05	0.07	0.06	S
ɔ:	5	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	36	0.00	0.00	0.00	0.19	0.11	0.14	M
ɛ	6	0.00	0.00	0.00	0.00	0.00	0.00	-
ɛi	3	0.33	0.03	0.06	0.00	0.00	0.00	S
ʏ	3	0.00	0.00	0.00	0.00	0.00	0.00	-
ɪ	16	0.12	0.20	0.15	0.00	0.00	0.00	S
ʊ	12	0.17	0.14	0.15	0.00	0.00	0.00	S
TOTAL	315	0.08	0.08	0.05	0.03	0.02	0.03	S

Phoneme Clustering Analysis - P28

Hierarchical Clustering (Ward linkage)

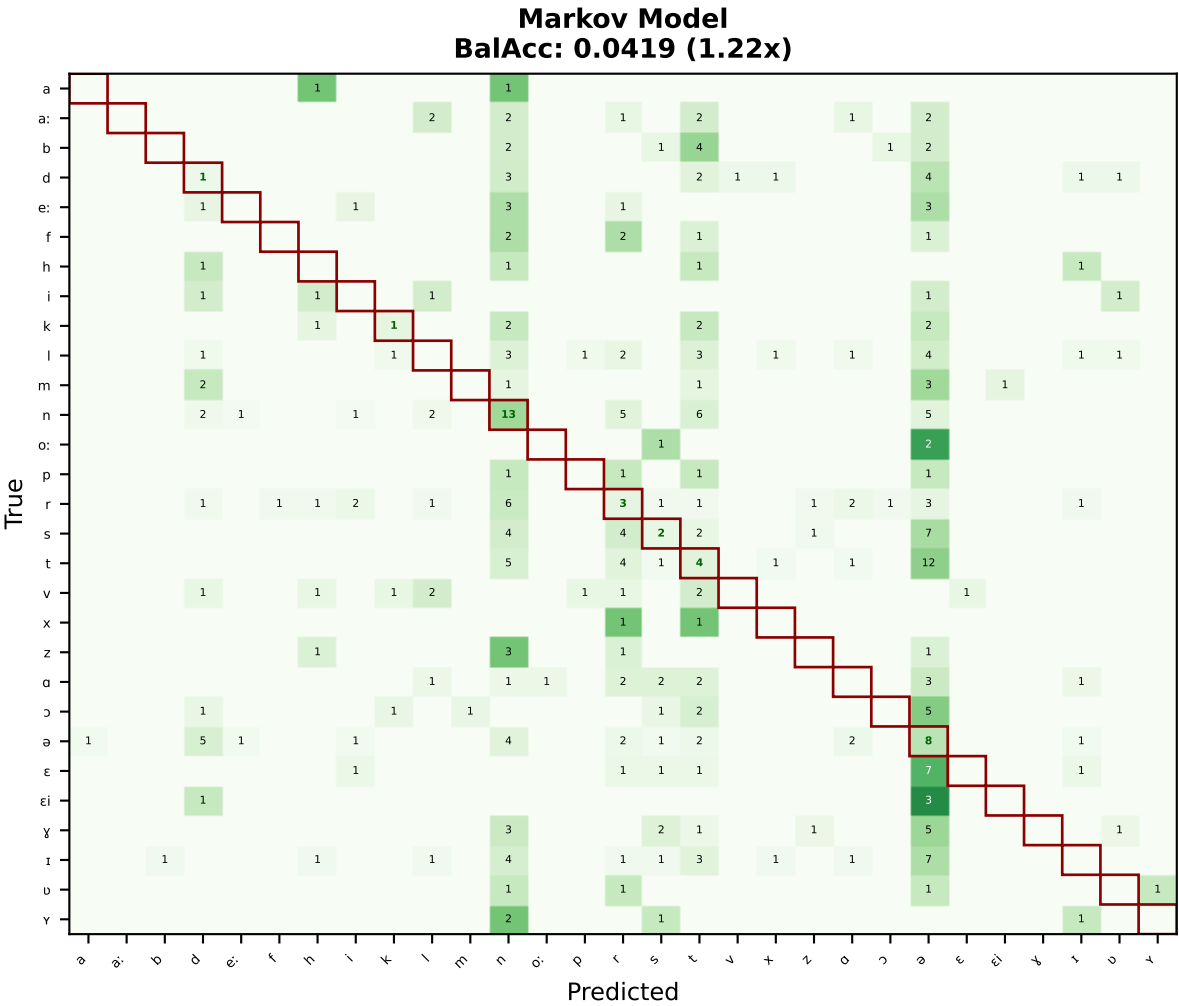
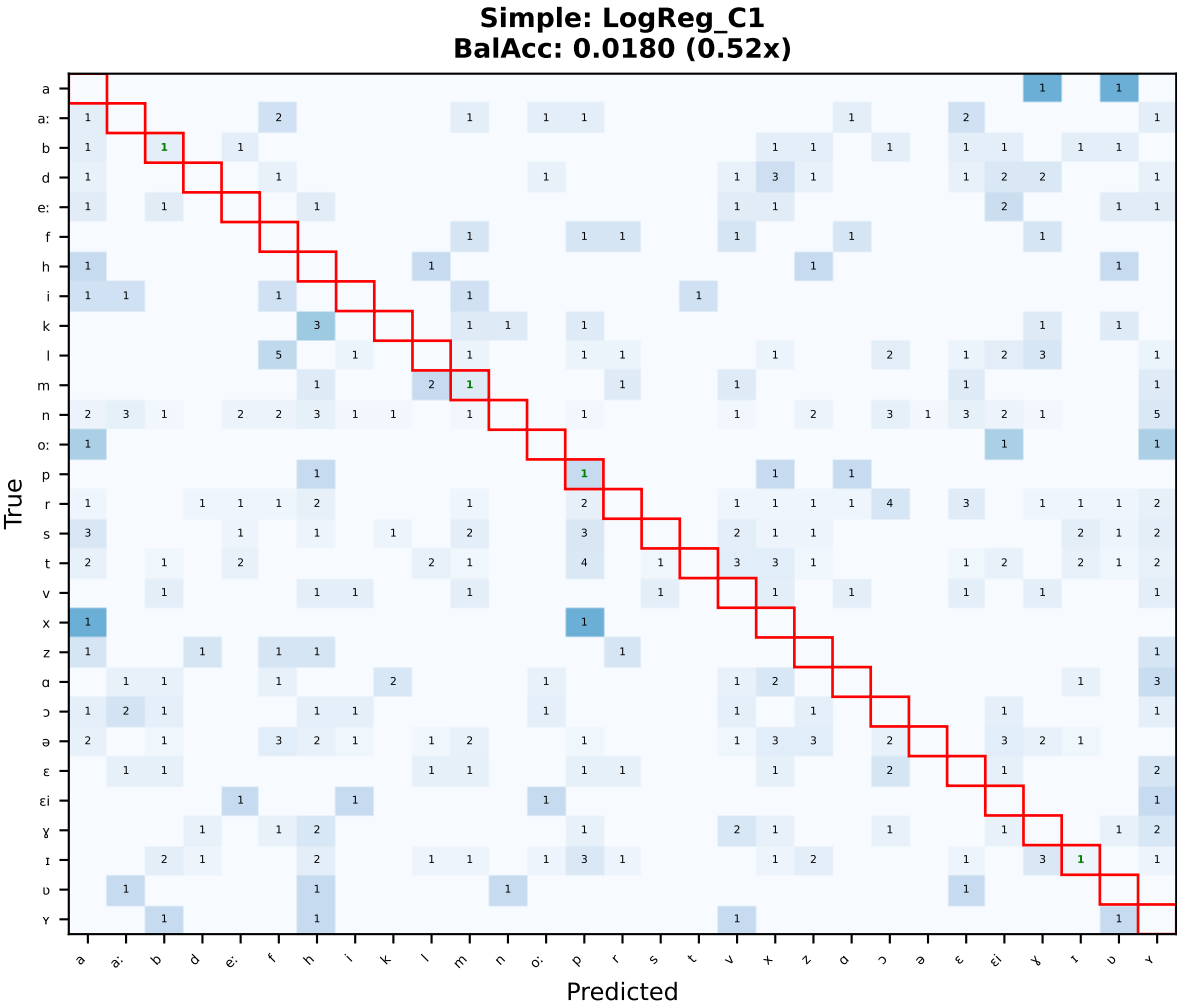


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	2	71	Plosive	p, ei
Group 2	1	10	Short Vowel	a
Group 3	4	240	Long Vowel	e:, i, r, v
Group 4	9	730	Short Vowel	a:, d, k, l, s, a, ə, ε, ...
Group 5	9	577	Short Vowel	b, h, m, n, o:, t, ɒ, ʌ, ...

Phonetic Categories

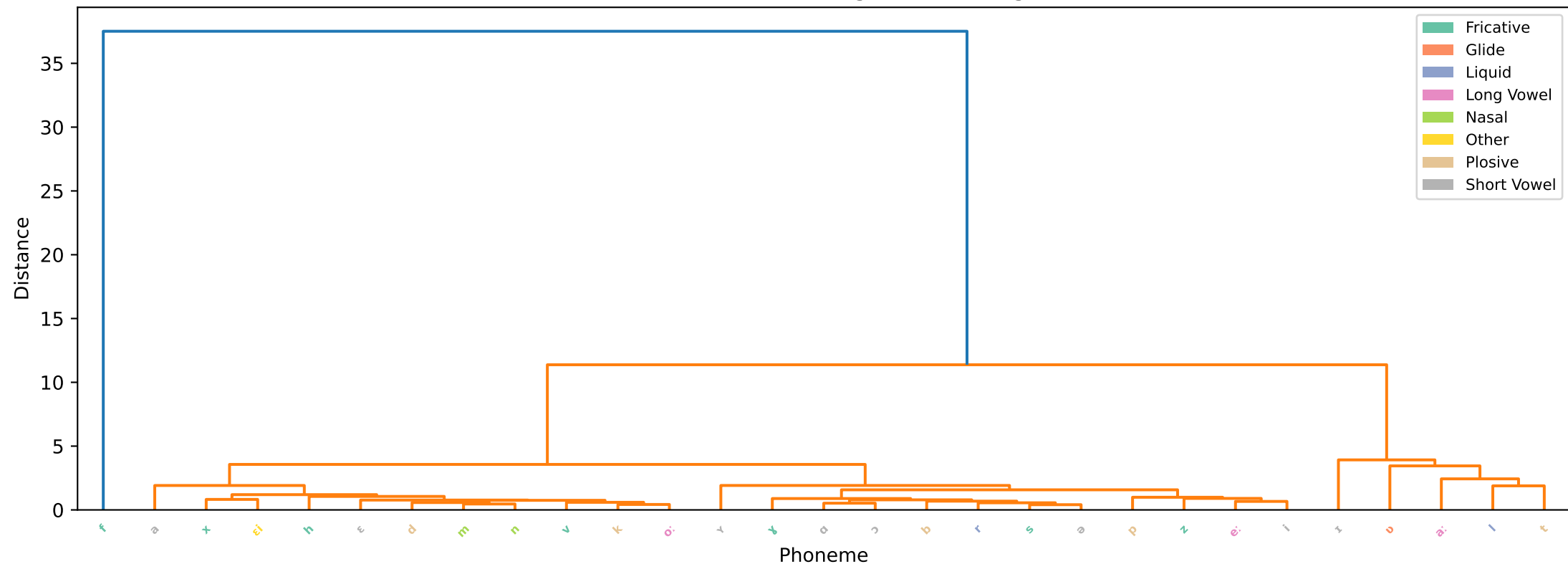
Category	N	Samples	Phonemes
Fricative	4	215	h, s, v, ʒ
Glide	1	34	ʊ
Liquid	2	193	l, r
Long Vowel	3	115	a:, e:, o:
Nasal	2	189	m, n
Other	1	36	εi
Plosive	5	368	b, d, k, p, t
Short Vowel	7	478	a, i, ɑ, ɔ, ə, ε, ɪ



Per-Phoneme Comparison (S=Simple, M=Markov)									
Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win	
a	2	0.00	0.00	0.00	0.00	0.00	0.00	-	
a:	10	0.00	0.00	0.00	0.00	0.00	0.00	-	
b	10	0.10	0.08	0.09	0.00	0.00	0.00	S	
d	14	0.00	0.00	0.00	0.07	0.06	0.06	M	
e:	9	0.00	0.00	0.00	0.00	0.00	0.00	-	
f	6	0.00	0.00	0.00	0.00	0.00	0.00	-	
h	4	0.00	0.00	0.00	0.00	0.00	0.00	-	
i	5	0.00	0.00	0.00	0.00	0.00	0.00	-	
k	8	0.00	0.00	0.00	0.12	0.25	0.17	M	
l	19	0.00	0.00	0.00	0.00	0.00	0.00	-	
m	8	0.12	0.06	0.08	0.00	0.00	0.00	S	
n	35	0.00	0.00	0.00	0.37	0.19	0.25	M	
o:	3	0.00	0.00	0.00	0.00	0.00	0.00	-	
p	4	0.25	0.05	0.08	0.00	0.00	0.00	S	
r	25	0.00	0.00	0.00	0.12	0.09	0.10	M	
s	20	0.00	0.00	0.00	0.10	0.13	0.11	M	
t	28	0.00	0.00	0.00	0.14	0.09	0.11	M	
v	10	0.00	0.00	0.00	0.00	0.00	0.00	-	
x	2	0.00	0.00	0.00	0.00	0.00	0.00	-	
z	6	0.00	0.00	0.00	0.00	0.00	0.00	-	
ɑ	13	0.00	0.00	0.00	0.00	0.00	0.00	-	
ɔ	11	0.00	0.00	0.00	0.00	0.00	0.00	-	
ə	28	0.00	0.00	0.00	0.29	0.09	0.13	M	
ɛ	12	0.00	0.00	0.00	0.00	0.00	0.00	-	
ɛi	4	0.00	0.00	0.00	0.00	0.00	0.00	-	
ʏ	13	0.00	0.00	0.00	0.00	0.00	0.00	-	
ɪ	21	0.05	0.11	0.07	0.00	0.00	0.00	S	
ʊ	4	0.00	0.00	0.00	0.00	0.00	0.00	-	
ʏ	4	0.00	0.00	0.00	0.00	0.00	0.00	-	
TOTAL	338	0.02	0.01	0.01	0.04	0.03	0.03	M	

Phoneme Clustering Analysis - P29

Hierarchical Clustering (Ward linkage)

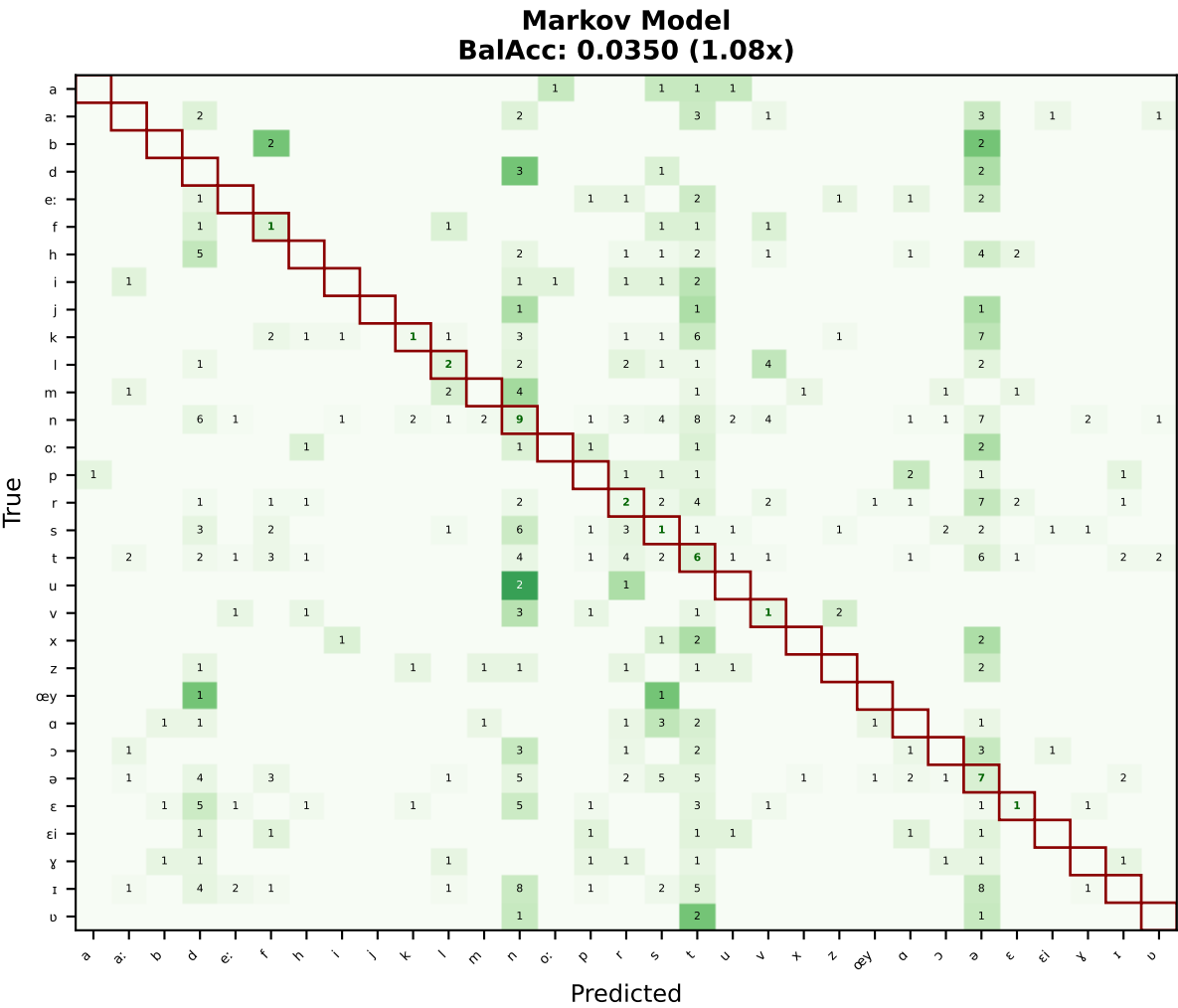
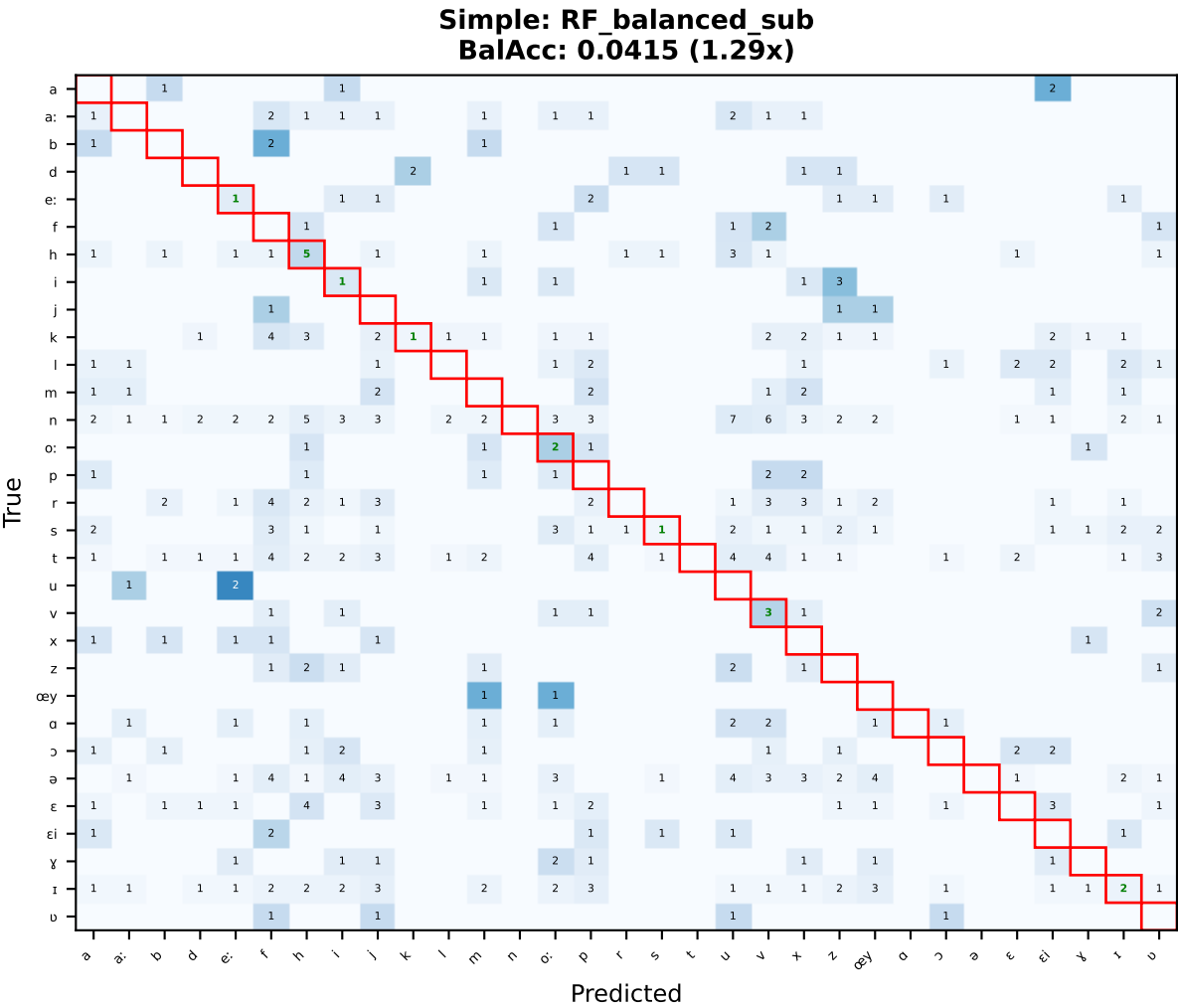


Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	11	733	Fricative	a, d, h, k, m, n, p:, v, ...
Group 2	12	889	Short Vowel	b, e:, i, p, r, s, z, a, ...
Group 3	4	368	Long Vowel	a:, l, t, ʊ
Group 4	1	65	Short Vowel	ɪ
Group 5	1	22	Fricative	f

Phonetic Categories

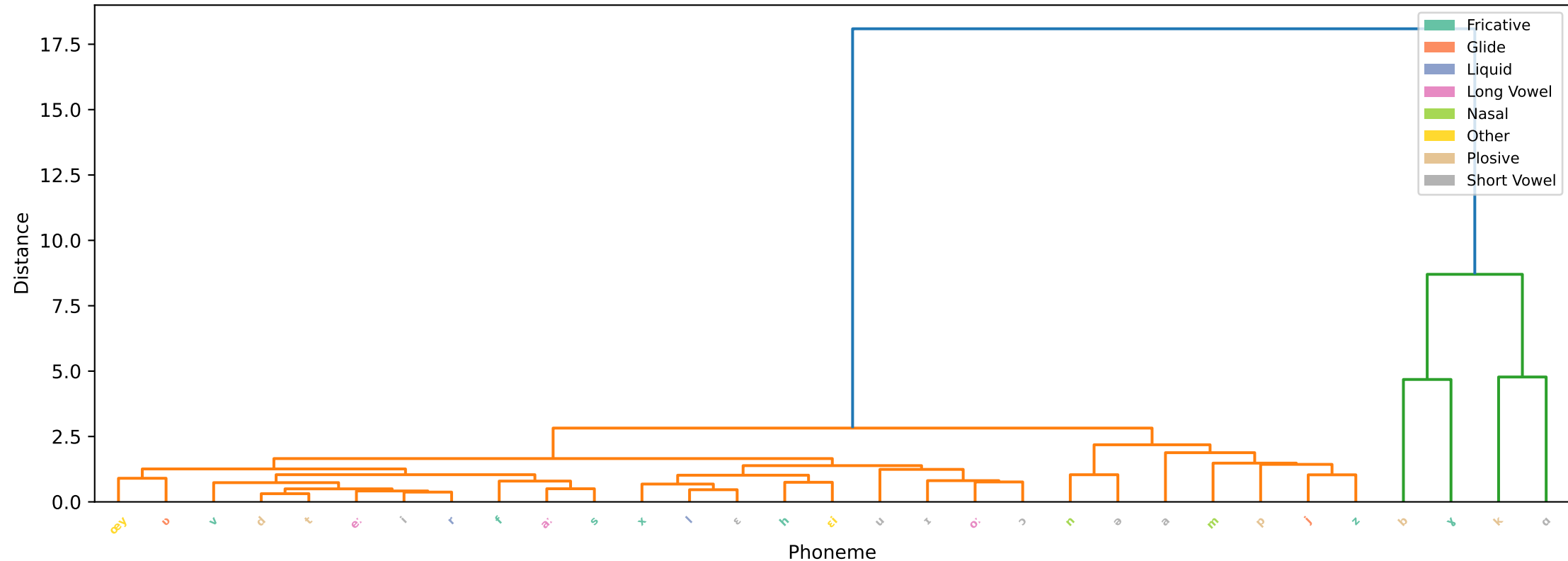
Category	N	Samples	Phonemes
Fricative	7	328	f, h, s, v, x, z, ʃ
Glide	1	51	ʊ
Liquid	2	238	l, r
Long Vowel	3	124	a:, e:, o:
Nasal	2	267	m, n
Other	1	36	ɛi
Plosive	5	420	b, d, k, p, t
Short Vowel	8	613	a, i, ʌ, ɔ, ə, ɛ, ɪ, ...



Phoneme	N	S-Rec	S-Prec	S-F1	M-Rec	M-Prec	M-F1	Win
a	4	0.00	0.00	0.00	0.00	0.00	0.00	-
a:	13	0.00	0.00	0.00	0.00	0.00	0.00	-
b	4	0.00	0.00	0.00	0.00	0.00	0.00	-
d	6	0.00	0.00	0.00	0.00	0.00	0.00	-
e:	9	0.11	0.07	0.09	0.00	0.00	0.00	S
f	6	0.00	0.00	0.00	0.17	0.06	0.09	M
h	19	0.26	0.15	0.19	0.00	0.00	0.00	S
i	7	0.14	0.05	0.07	0.00	0.00	0.00	S
j	3	0.00	0.00	0.00	0.00	0.00	0.00	-
k	25	0.04	0.33	0.07	0.04	0.20	0.07	S
l	15	0.00	0.00	0.00	0.13	0.18	0.15	M
m	11	0.00	0.00	0.00	0.00	0.00	0.00	-
n	56	0.00	0.00	0.00	0.16	0.13	0.15	M
o:	6	0.33	0.08	0.13	0.00	0.00	0.00	S
p	8	0.00	0.00	0.00	0.00	0.00	0.00	-
r	27	0.00	0.00	0.00	0.07	0.08	0.08	M
s	26	0.04	0.17	0.06	0.04	0.03	0.04	S
t	40	0.00	0.00	0.00	0.15	0.09	0.11	M
u	3	0.00	0.00	0.00	0.00	0.00	0.00	-
v	10	0.30	0.09	0.14	0.10	0.06	0.08	S
x	6	0.00	0.00	0.00	0.00	0.00	0.00	-
z	9	0.00	0.00	0.00	0.00	0.00	0.00	-
œy	2	0.00	0.00	0.00	0.00	0.00	0.00	-
ɑ	11	0.00	0.00	0.00	0.00	0.00	0.00	-
ɔ	12	0.00	0.00	0.00	0.00	0.00	0.00	-
ə	40	0.00	0.00	0.00	0.17	0.09	0.12	M
ɛ	22	0.00	0.00	0.00	0.05	0.14	0.07	M
ɛi	7	0.00	0.00	0.00	0.00	0.00	0.00	-
ɣ	9	0.00	0.00	0.00	0.00	0.00	0.00	-
ɪ	34	0.06	0.12	0.08	0.00	0.00	0.00	S
ʊ	4	0.00	0.00	0.00	0.00	0.00	0.00	-
TOTAL	454	0.04	0.03	0.03	0.03	0.03	0.03	S

Phoneme Clustering Analysis - P30

Hierarchical Clustering (Ward linkage)



Suggested Groupings (5 clusters)

Group	N	Samples	Type	Phonemes
Group 1	27	1913	Short Vowel	a, a:, d, e:, f, h, j,...
Group 2	1	39	Plosive	b
Group 3	1	47	Fricative	ɣ
Group 4	1	60	Plosive	k
Group 5	1	97	Short Vowel	ɑ

Phonetic Categories

Category	N	Samples	Phonemes
Fricative	7	346	f, h, s, v, x, z, ɣ
Glide	2	67	j, ʊ
Liquid	2	207	l, r
Long Vowel	3	124	a:, e:, ɔ:
Nasal	2	267	m, n
Other	2	49	æy, ɛi
Plosive	5	475	b, d, k, p, t
Short Vowel	8	621	a, i, u, ɑ, ɔ, ə, ɛ,...